

# ENVIRONMENTAL STUDIES (EVS) COMPULSORY

## Ability Enhancement Compulsory Course (AECC-2) — CBCS Degree Examination

★ THE ALL-IN-ONE 24-HOUR EMERGENCY MASTER COMPENDIUM & CRASH PREPARATION GUIDE ★

|                                            |                                 |                                            |
|--------------------------------------------|---------------------------------|--------------------------------------------|
| Total Marks: 100 Marks (50 MCQs × 2 Marks) | Duration: 2 Hours (120 Minutes) | Negative Marking: <b>NONE (0% Penalty)</b> |
| Exam Pattern: Pure OMR Multiple Choice     | Target Score: 90+ / 100 Marks   | Coverage: Units 1 to 7 + 100% Solved PYQs  |

**CRITICAL ZERO-HOUR SUMMARY & GOLDEN DIRECTIVE:**

This master compendium has been compiled directly from your official **UGC CBCS AECC-2 Syllabus**, **Erach Bharucha's official UGC Textbook**, the **Loreto College Question Bank**, and **three consecutive years of actual university degree examination papers (2024, 2023, and 2022)**.

**THE NUMBER ONE STRATEGIC IMPERATIVE:** Instruction 8 of your Question Booklet explicitly dictates: 'There is no Negative mark.' Therefore, **leaving any question blank is fatal**. You **MUST** shade all 50 OMR circles. Furthermore, empirical analysis reveals that over **80% to 90% of marks** are concentrated in just 4 units: **Unit 5 (Pollution & Waste - 42%)**, **Unit 3 (Natural Resources & Energy - 17%)**, **Unit 6 (Environmental Laws, Climate & Ozone - 17%)**, and **Unit 4 (Biodiversity & Conservation - 14%)**. Follow this guide systematically tonight to guarantee an outstanding score tomorrow.

## SECTION 1: EXAM ARCHITECTURE & STRATEGIC BLUEPRINT

### 1.1 Empirical Unit-Wise Weightage Breakdown

Based on deep statistical analysis of 150+ actual exam questions across 2022, 2023, and 2024, the marks and question distribution across the official UGC syllabus is mapped below:

| Unit # | Syllabus Unit Title                                     | Average Weightage | Expected Qs   | Expected Marks | Preparation Priority |
|--------|---------------------------------------------------------|-------------------|---------------|----------------|----------------------|
| Unit 5 | Environmental Pollution & Solid/Liquid Waste Management | 42%               | 21 Questions  | 42 Marks       | CRITICAL MUST-MASTER |
| Unit 3 | Natural Resources (Water, Forests, Energy, Mega-Dams)   | 17%               | 8-9 Questions | 18 Marks       | HIGH PRIORITY        |
| Unit 6 | Environmental Policies, Legislation, Climate & Ozone    | 17%               | 8-9 Questions | 18 Marks       | HIGH PRIORITY        |
| Unit 4 | Biodiversity & In-situ / Ex-situ Conservation           | 14%               | 7 Questions   | 14 Marks       | HIGH PRIORITY        |
| Unit 1 | Introduction to EVS & Sustainable Development           | 4%                | 2 Questions   | 4 Marks        | Moderate Priority    |
| Unit 2 | Ecosystems (Food Chains, Food Webs, Energy Flow)        | 3%                | 1-2 Questions | 3 Marks        | Moderate Priority    |
| Unit 7 | Human Communities, Movements & Disaster Management      | 3%                | 1-2 Questions | 3 Marks        | Moderate Priority    |

## 1.2 The 24-Hour Zero-to-Hero Emergency Study Schedule

| Time Slot                    | Phase               | Core Target & Execution Strategy                                                                                                               |
|------------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Night: 08:00 PM - 11:00 PM   | Core Blast (Unit 5) | Master Unit 5 (Pollution & Waste). Memorize DO, BOD/COD, Alum, Activated Sludge, Hardness, Incineration, Bhopal Gas (MIC), and Disease toxins. |
| Night: 11:15 PM - 01:30 AM   | Policy & Resources  | Master Unit 6 (Laws, Dates, Montreal/Kyoto Protocols, Ozone DU) and Unit 3 (Forests, Dams, Energy fuels, Jatropha, Hydrogen, CNG).             |
| Night: 01:30 AM - 02:30 AM   | Biodiversity Drill  | Master Unit 4 (4 Indian Hotspots, In-situ vs Ex-situ, Kaziranga, Corbett, Red Data Book IUCN, DDT biomagnification).                           |
| Morning: 06:30 AM - 08:30 AM | PYQ Speed-Run       | Read through the 100% Solved 2024, 2023, and 2022 Question Papers in Sections 4, 5, and 6.                                                     |
| Final Hour: Before Exam      | Rapid Cheat-Sheets  | Revise Section 2 Cheat Tables (Acts/Years, Days, Toxins, Parks). Remember: Attempt all 50 questions!                                           |

SECTION 2: HIGH-YIELD MASTER REVISION CHEAT-SHEETS

Table 2.1: Key Indian Environmental Legislation & Years (1971–2010)

| Act / Policy Name                             | Enacted / Amended   | Core Purpose & Statutory Highlights                                                                                                     |
|-----------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Wildlife Protection Act                       | 1972                | Constitutes National Parks, Sanctuaries; prohibits hunting; Schedules I-VI for endangered flora/fauna.                                  |
| Water (Prevention & Control of Pollution) Act | 1974                | Establishes Central & State Pollution Control Boards (CPCB/SPCB); regulates industrial effluents into water bodies.                     |
| Forest (Conservation) Act                     | 1980 (Amended 1988) | Restricts de-reservation of forests or diversion of forest land for non-forest purposes without Central Govt approval.                  |
| Air (Prevention & Control of Pollution) Act   | 1981 (Amended 1987) | Regulates industrial and vehicular air emissions; 1987 amendment formally included Noise as an air pollutant.                           |
| Environment (Protection) Act (EPA)            | 1986                | Umbrella legislation enacted after Bhopal Disaster (Art. 253). Mandates EIA (Environmental Impact Assessment) and nationwide standards. |
| Public Liability Insurance Act                | 1991                | Mandatory insurance providing immediate relief to persons affected by accidents while handling hazardous substances.                    |
| Biological Diversity Act                      | 2002                | Decentralized conservation via National Biodiversity Authority (NBA), State Boards, and local BMCs; implements CBD 1992.                |
| National Green Tribunal (NGT) Act             | 2010                | Specialized judicial body under Article 21 for speedy disposal (within 6 months) of civil environmental disputes.                       |

Table 2.2: Landmark Global Environmental Treaties, Protocols & Summits

| Treaty / Convention                     | Year & Location      | Key Goal & Historic Significance                                                                                          |
|-----------------------------------------|----------------------|---------------------------------------------------------------------------------------------------------------------------|
| Ramsar Convention                       | 1971 (Ramsar, Iran)  | Intergovernmental treaty for conservation and sustainable utilization of global wetlands.                                 |
| Stockholm UN Conference                 | 1972 (Sweden)        | First major UN global environmental summit; established UNEP; led to World Environment Day (June 5).                      |
| Brundtland Report ('Our Common Future') | 1987 (WCED)          | Formally defined 'Sustainable Development' as meeting present needs without compromising future generations.              |
| Montreal Protocol                       | 1987 (Canada)        | Legally binding treaty to phase out Ozone Depleting Substances (ODS) including CFCs, halons, and carbon tetrachloride.    |
| Rio Earth Summit (UNCED)                | 1992 (Rio, Brazil)   | Landmark conference produced Agenda 21, Rio Declaration, CBD (Convention on Biological Diversity), and UNFCCC.            |
| Kyoto Protocol                          | 1997 (Kyoto, Japan)  | Under UNFCCC; legally binding emission reduction targets for Annex I nations on 6 GHGs; introduced Carbon Trading/CDM.    |
| Johannesburg Earth Summit (Rio+10)      | 2002 (South Africa)  | Focus on sustainable development, poverty eradication, sanitation, and clean water access.                                |
| Paris Climate Agreement (COP 21)        | 2015 (Paris, France) | Legally binding global treaty to limit global warming to well below 2.0°C (preferably 1.5°C) above pre-industrial levels. |

**Table 2.3: Important Environmental Calendar Days & Observances**

| Date         | Observance             | Key Focus / Background                                                                     |
|--------------|------------------------|--------------------------------------------------------------------------------------------|
| February 2   | World Wetlands Day     | Commemorates the signing of the Ramsar Convention on Wetlands (1971).                      |
| March 21     | World Forest Day       | Promotes awareness of the importance of all types of forests and woodlands.                |
| March 22     | World Water Day        | Focuses on global freshwater crisis, conservation, and SDG 6 (Clean Water and Sanitation). |
| April 22     | Earth Day              | First observed in 1970; global civic movement for environmental protection.                |
| June 5       | World Environment Day  | Flagship UN environmental day; marks the opening of Stockholm Conference (1972).           |
| July 1-7     | Van Mahotsav Week      | Annual Indian tree-planting movement initiated in 1950 by K.M. Munshi.                     |
| September 16 | World Ozone Day        | Commemorates the signing of the Montreal Protocol (1987) on substances that deplete ozone. |
| October 2-8  | National Wildlife Week | Annual Indian observance dedicated to the preservation of India's faunal biodiversity.     |

**Table 2.4: Diseases, Toxins & Environmental Health Hazards Matrix**

| Disease / Health Hazard     | Causative Agent / Toxin              | Transmission Vector / Mechanism & Key Symptoms                                                               |
|-----------------------------|--------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Minamata Disease            | Methylmercury (Hg)                   | Bioaccumulated through consumption of contaminated marine fish; neurological damage.                         |
| Itai-Itai Disease           | Cadmium (Cd)                         | Accumulated via industrial wastewater in paddy fields; osteomalacia, severe joint & bone pain.               |
| Blackfoot Disease           | Inorganic Arsenic (As)               | Groundwater contamination in alluvial aquifers; peripheral vascular gangrene, hyperkeratosis.                |
| Skeletal & Dental Fluorosis | Excess Fluoride (F > 1.5 mg/L)       | Groundwater leaching; causes mottling of teeth enamel, bone deformities, calcification of ligaments.         |
| Blue Baby Syndrome          | Nitrates (NO <sub>3</sub> > 45 mg/L) | Methaemoglobinaemia in infants; nitrate reduces blood oxygen-carrying capacity causing cyanosis.             |
| Cholera, Typhoid, Dysentery | Pathogenic Bacteria / Viruses        | Direct waterborne fecal-oral transmission via contaminated domestic drinking water supplies.                 |
| Malaria                     | Plasmodium vivax / falciparum        | Vector-borne (NOT waterborne directly); transmitted by female Anopheles mosquito breeding in stagnant water. |
| Dengue & Chikungunya        | Flavivirus / Arbovirus               | Vector-borne; transmitted by daytime biting Aedes aegypti mosquito breeding in clean stagnant water.         |

**Table 2.5: Biodiversity, Hotspots & Conservation Matrix**

| Classification / Concept | Key Details / Numbers            | Crucial Examples & Exam Facts                                                                                       |
|--------------------------|----------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Global Hotspots          | 36 Total in World                | Identified by Norman Myers: Must have $\geq$ 1,500 endemic vascular plants and $\geq$ 70% habitat loss.             |
| Indian Hotspots          | 4 Hotspots cover India           | 1) Western Ghats, 2) Eastern Himalayas, 3) Indo-Burma, 4) Sundaland (includes Nicobar).                             |
| Oldest National Park     | Jim Corbett National Park        | Established 1936 (as Hailey NP) in Nainital/Pauri Garhwal, Uttarakhand; first Project Tiger park.                   |
| One-Horned Rhinoceros    | Kaziranga National Park          | Located in Assam on Brahmaputra floodplain; holds world's largest population of Greater One-Horned Rhinos.          |
| Asiatic Lions            | Gir National Park                | Located in Gujarat; sole remaining natural habitat of the Asiatic Lion (Panthera leo persica).                      |
| In-Situ Conservation     | On-site protected habitats       | National Parks, Wildlife Sanctuaries, Biosphere Reserves (e.g. Sundarbans, Nanda Devi), Sacred Groves.              |
| Ex-Situ Conservation     | Off-site artificial preservation | Botanical Gardens, Zoological Parks, Seed Banks, Cryopreservation (-196°C in liquid N <sub>2</sub> ), Gene Banks.   |
| IUCN Red Data Book       | Published by IUCN                | Comprehensive global status inventory categorizing species: Extinct, Critically Endangered, Endangered, Vulnerable. |

**Table 2.6: Water, Wastewater & Environmental Engineering Parameters**

| Parameter / Process             | Standard / Value                                              | Significance & Mechanism                                                                                                                                   |
|---------------------------------|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dissolved Oxygen (DO)           | Minimum 4 to 6 mg/L                                           | Essential for aerobic aquatic life. DO drops rapidly during organic pollution decomposition.                                                               |
| Biochemical Oxygen Demand (BOD) | < 3 mg/L (clean)   > 10 mg/L (polluted)                       | Oxygen needed by aerobic microbes to degrade organic matter in 5 days at 20°C (BOD <sub>5</sub> ).                                                         |
| Chemical Oxygen Demand (COD)    | Always > BOD                                                  | Total oxygen required to chemically oxidize all organic matter (both biodegradable and non-biodegradable).                                                 |
| Eutrophication                  | Excess Nitrates & Phosphates                                  | Nutrient loading leads to Algal Blooms; Phosphorus is the primary limiting nutrient in freshwater.                                                         |
| Coagulation / Flocculation      | Alum: $\text{Al}_2(\text{SO}_4)_3 \cdot 18\text{H}_2\text{O}$ | Reacts with water alkalinity to precipitate gelatinous Aluminium Hydroxide $[\text{Al}(\text{OH})_3]$ that traps colloids.                                 |
| Biological Secondary Treatment  | Activated Sludge Process                                      | Aerobic mixed microbial suspension converts dissolved organic waste into settleable biological floc.                                                       |
| Disinfection                    | Chlorine / UV Radiation                                       | Destroys pathogens and enteric microorganisms in treated sewage prior to final environmental discharge.                                                    |
| Water Hardness                  | $\text{Ca}^{2+}$ and $\text{Mg}^{2+}$ salts                   | Permanent: Sulphates & Chlorides ( $\text{MgSO}_4$ , $\text{CaCl}_2$ ). Temporary: Bicarbonates $[\text{Ca}(\text{HCO}_3)_2, \text{Mg}(\text{HCO}_3)_2]$ . |

## SECTION 3: COMPREHENSIVE UNIT-BY-UNIT THEORY NOTES

These notes cover 100% of the concepts, mechanisms, and factual questions in the UGC CBCS curriculum, meticulously extracted from Erach Bharucha's textbook and university examination syllabi.

### Unit 1: Introduction to Environmental Studies & Sustainable Development

**Exam Weightage: 4% of Exam | Approx 2 Questions (4 Marks)**

- **Multidisciplinary Nature of EVS:** Environmental Studies bridges physical sciences (physics, chemistry, meteorology), biological sciences (ecology, botany, zoology), engineering (civil, chemical, waste management), and social sciences (economics, sociology, law, ethics). It provides a holistic approach to understanding human-environment interactions.
- **Scope and Practical Importance:** EVS equips citizens and professionals to manage natural resource depletion, control environmental pollution, conserve endangered biological diversity, and implement sustainable industrial and urban development.
- **Concept of Sustainability & Sustainable Development:** First comprehensively defined in the 1987 Brundtland Commission Report ('Our Common Future', published by the UN World Commission on Environment and Development chaired by Gro Harlem Brundtland): 'Sustainable development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs'.
- **Core Pillars of Sustainable Development:** It requires the balanced integration of three pillars: 1) Economic Growth, 2) Social Equity & Human Welfare, and 3) Environmental Protection and Ecological Integrity.
- **Inter-generational and Intra-generational Equity:** Inter-generational equity holds that the present generation holds Earth's resources in trust for future generations. Intra-generational equity requires equitable sharing of environmental resources and clean environments among all social groups within the current generation.
- **The 3 R's Hierarchy of Waste and Resource Management:** 1) Reduce: Minimize waste generation at source (most preferred); 2) Reuse: Use items repeatedly without processing; 3) Recycle: Collect, process, and remanufacture waste materials into new products (e.g., aluminium remelted indefinitely with 95% energy savings).

### Unit 2: Ecosystems — Structure, Function & Energy Dynamics

**Exam Weightage: 3% of Exam | Approx 1-2 Questions (2-4 Marks)**

- **Definition of Ecosystem:** The term 'Ecosystem' was coined in 1935 by British ecologist Sir Arthur George Tansley. An ecosystem is a functional, self-sustaining ecological unit consisting of biotic communities interacting continuously with their abiotic physical-chemical environment.
- **Structural Components of an Ecosystem:** 1) Abiotic Components: Sunlight, temperature, precipitation, atmospheric gases (O<sub>2</sub>, CO<sub>2</sub>), water, soil minerals, and nutrients (N, P, K); 2) Biotic Components: Autotrophs (producers), Heterotrophs (consumers), and Saprotrophs (decomposers).
- **Trophic Levels and Biotic Classifications:**
  - Trophic Level 1 (T1): Primary Producers (photoautotrophic green plants, phytoplanktons, cyanobacteria).
  - Trophic Level 2 (T2): Primary Consumers (herbivores: deer, cattle, zooplankton).
  - Trophic Level 3 (T3): Secondary Consumers (primary carnivores: frogs, small fish).
  - Trophic Level 4 (T4): Tertiary Consumers (top predators: tigers, eagles, sharks).
  - Decomposers (Micro-consumers): Fungi and bacteria that chemically break down dead organic matter (detritus), recycling mineral nutrients back to the abiotic soil/water pool.
- **Energy Flow in Ecosystems:** Energy flow through an ecosystem is strictly UNIDIRECTIONAL (governed by the First and Second Laws of Thermodynamics). Solar radiant energy is fixed by producers and transferred up the food chain, with substantial energy lost at each trophic step as metabolic heat. According to Raymond Lindeman's 10% Ecological Efficiency Law (1942), only approximately 10% of the energy stored as organic biomass at one trophic level is transferred and assimilated into biomass at the subsequent higher trophic level.
- **Food Chains and Food Webs:**
  - Grazing Food Chain: Originates with living autotrophic green plants → herbivores → carnivores (e.g., Grass → Grasshopper → Frog → Snake → Hawk).
  - Detritus Food Chain: Originates with non-living dead organic matter (plant litter, animal carcasses) → detritivores (blowfly maggots, earthworms, woodlice) → predators (e.g., Dead Animal → Blowfly Maggots → Frog → Snake). In terrestrial ecosystems, the majority of energy flows through detritus pathways.
  - Food Web: A complex, highly interconnected network of multiple overlapping food chains within an ecological community. Higher

food web complexity imparts greater ecological stability and resilience against environmental disturbances.

- **Ecological Pyramids:** Conceptualized by Charles Elton (1927):
  - Pyramid of Energy: ALWAYS upright in all ecosystems without exception, because energy is irreversibly lost as metabolic heat at each transfer level.
  - Pyramid of Biomass: Upright in terrestrial forest/grassland ecosystems; inverted in aquatic marine/pond ecosystems (where a small instantaneous standing biomass of short-lived phytoplankton supports a much larger standing biomass of long-lived zooplankton and fish).
  - Pyramid of Numbers: Upright in grasslands; inverted or spindle-shaped in a tree parasitic system (one large tree supports thousands of herbivorous insects and avian parasites).
- **Ecological Succession:** The orderly, predictable, directional process of community change over time leading to a stable, self-perpetuating climax community.
  - Primary Succession: Initiates on previously uninhabited, sterile substrates devoid of pre-existing soil (e.g., cooled volcanic lava, bare rock, newly exposed sand dunes). First colonizers are 'Pioneer Species' (e.g., lichens on bare rock that secrete acids to weather stone).
  - Secondary Succession: Initiates on sites where an established ecological community was disturbed or destroyed, but where organic soil and seed banks remain intact (e.g., abandoned farmland, burnt forests, floodplains). Secondary succession proceeds much faster than primary succession.
- **Aquatic Ecosystem Classifications:**
  - Lotic Freshwater: Running, continuously flowing freshwaters (rivers, mountain streams, rapids). High dissolved oxygen, adaptations for current resistance.
  - Lentic Freshwater: Standing, calm, motionless waters (lakes, ponds, swamps, bogs). Exhibit thermal stratification.
  - Plankton: Microscopic organisms suspended in the water column drifting with water currents (Phytoplankton: diatoms, green algae; Zooplankton: rotifers, copepods).
  - Nekton: Actively swimming aquatic organisms capable of moving against currents (fish, amphibians, aquatic mammals).
  - Benthos: Organisms residing on or burrowed within the bottom sediments (molluscs, crustaceans, benthic worms).
  - Estuaries and Deltas: Semi-enclosed coastal bodies of water where freshwater from rivers mixes with saline ocean water, creating dynamic brackish water ecosystems characterized by exceptionally high biological productivity and extensive mangrove forests.

### Unit 3: Natural Resources — Renewable, Non-Renewable & Energy Transitions

**Exam Weightage: 17% of Exam | Approx 8-10 Questions (16-20 Marks)**

- **Classification of Natural Resources:**
  - Renewable Inexhaustible: Abundant, perpetual energy flows not depleted by human use (Solar radiation, Wind energy, Tidal power, Geothermal heat).
  - Renewable Exhaustible: Biologically regenerating resources that can be depleted if harvesting rate exceeds natural regeneration rate (Forests, Wild fauna, Groundwater aquifers, Fertile topsoils).
  - Non-Renewable (Exhaustible): Finite physical stock formed over geological epochs that cannot regenerate on human timescales (Fossil fuels: coal, crude petroleum, natural gas; Metallic minerals: iron, copper, bauxite; Non-metallic minerals: limestone, granite, marble).
- **Forest Resources & Deforestation:**
  - Ecological Functions: Climate regulation, carbon sequestration, watershed protection, soil retention, aesthetic and spiritual values.
  - Direct vs Indirect Uses (Erach Bharucha Textbook, Page 96): Direct uses include gathered fruits, edible roots, direct medicinal herbs, fuelwood, timber for huts, and grass cut for livestock grazing. Indirect uses include commercially processed gums and resins, industrial pharmaceuticals, synthetic chemicals, and paper manufactured from bamboo/softwoods.
  - Drivers of Deforestation: Shifting (Jhum) cultivation, agricultural conversion, commercial logging, mining, and mega-infrastructure (dams and highways).
  - Policy Mandate: India's National Forest Policy (1988) targets an optimal 33% total geographical forest/tree cover across the nation (67% in mountainous hill tracts to prevent soil erosion and landslides, 20% in agricultural plains).
- **Water Resources, Overexploitation & Drought:**
  - Hydrological Cycle: The continuous solar-driven recycling of planetary water between atmosphere, lithosphere, and oceans via evaporation, transpiration, condensation, precipitation, and runoff.
  - Overexploitation Crisis: Widespread groundwater over-pumping via millions of subsidized deep agricultural borewells causes irreversible aquifer depletion, land subsidence, and salinity ingress.
  - Rainwater Harvesting (RWH): Capturing surface precipitation to artificially recharge depleted subterranean aquifers. Tamil Nadu was the first Indian state to make rooftop RWH legally mandatory for all buildings (2001).
  - Drought: Initiated by 'Rain Failure' (monsoon deficit). IMD defines Moderate Drought when rainfall deficiency is 26% to 50%, and Severe Drought when deficit exceeds 50%, triggering widespread agricultural famine.
- **Dams — Multi-Purpose Benefits vs Socio-Ecological Disasters:**
  - Global Context: China and India together contain approximately 57% of all operational large dams worldwide.

- **Bhakra Nangal Dam:** Built across the Sutlej River in Himachal Pradesh; historically celebrated as India's largest concrete gravity dam.
- **Tehri Dam:** Located on the Bhagirathi River in Uttarakhand; India's tallest dam (260 m). Crucially, it sits in Seismic Zone V (active Himalayan fault zone), posing severe catastrophe risks.
- **Sardar Sarovar Dam:** Multipurpose terminal project on the Narmada River in Gujarat; generated intense national opposition (Narmada Bachao Andolan led by Medha Patkar and Baba Amte) due to the displacement of tens of thousands of tribal villagers without adequate rehabilitation.
- **Upstream Impacts:** Submergence of fertile river valleys and pristine forests, displacement of indigenous tribes, vector disease breeding.
- **Downstream Impacts:** Trapping of vital fertile river silt, riverbed erosion, loss of delta replenishment, altered estuarine salinity.
- **Energy Resources: Fossil vs Alternate Renewable Fuels:**
  - **Commercial Energy Mix:** Worldwide, crude oil (petroleum) is the largest single source of commercial energy consumption (~39%), followed by coal (~24%) and natural gas (~24%). In India, Coal is the predominant commercial energy source, accounting for ~55% of total commercial consumption.
  - **Rural Energy:** Fuelwood and dried animal dung cakes account for over 40% of primary energy in rural India.
  - **Nuclear Energy:** Tarapur Atomic Power Station (TAPS, Maharashtra, 1969) was India's first nuclear power station.
  - **Hydraulic Fracturing (Fracking):** High-pressure injection of water, sand, and chemical additives to fracture tight subterranean rock strata to extract shale oil and shale gas.
  - **CNG (Compressed Natural Gas):** Clean-burning automotive fuel primarily composed of methane (CH<sub>4</sub>); produces minimal particulate soot, lower CO, and reduced greenhouse gas emissions.
  - **Biofuels:** Renewable liquid transport fuels. Bioethanol and Biomethanol are the two primary alcohol biofuels. In India, Jatropha curcas non-edible seed oil is transesterified into commercial Biodiesel.
  - **Future Fuel:** Green Hydrogen (H<sub>2</sub>) fuel, when combusted in fuel cells or engines, releases only water vapor (H<sub>2</sub>O), offering the ultimate zero-emission transport energy carrier.

## Unit 4: Biodiversity & Conservation — Hotspots, Threats & Strategies

**Exam Weightage: 14% of Exam | Approx 7 Questions (14 Marks)**

- **Levels of Biological Diversity:**
  - **Genetic Diversity:** Diversity of genes within a single biological species (e.g., India has over 1,000 distinct cultivars of Mango and 50,000 varieties of Rice).
  - **Species Diversity:** The variety and richness of different species within a specific habitat or ecological community.
  - **Ecosystem Diversity:** The diversity of distinct habitats, biotic communities, and ecological processes (e.g., coral reefs, alpine meadows, mangrove estuaries).
- **India as a Mega-Biodiversity Nation:** India occupies only 2.4% of the world's total land area, yet accounts for nearly 7-8% of all recorded global species of flora and fauna, placing it among the 17 recognized megadiverse nations of the world.
- **Global and Indian Biodiversity Hotspots:**
  - **Criteria (Norman Myers):** A region must contain at least 1,500 species of vascular plants as endemics (> 0.5% of world's total) and have lost at least 70% of its original primary natural habitat.
  - **Global Total:** Exactly 36 Biodiversity Hotspots recognized worldwide.
  - **Indian Subcontinent:** 4 Hotspots extend across Indian territory:
    - 1) The Western Ghats (and Sri Lanka): Immense endemic floral, amphibian, and reptilian diversity.
    - 2) The Eastern Himalayas: Extends across Nepal, Bhutan, Northeast India (Arunachal Pradesh).
    - 3) Indo-Burma: Covers northeastern states south of the Brahmaputra, Assam, Manipur, Mizoram.
    - 4) Sundaland: Covers Nicobar Islands (extending through Malaysia, Indonesia, and Borneo).
- **Endemic, Endangered & Extinct Species:**
  - **Endemic Species:** Species geographically confined exclusively to a restricted, localized region and found nowhere else on Earth (e.g., Lion-tailed Macaque, Nilgiri Tahr in Western Ghats).
  - **Extinct Species:** When the very last surviving individual of a species dies worldwide (e.g., Dodo, Passenger Pigeon, Asiatic Cheetah in India declared extinct in 1952).
  - **IUCN Red Data Book:** Compiled by the International Union for Conservation of Nature (IUCN); categorizes species risk into Extinct (EX), Critically Endangered (CR - e.g., Great Indian Bustard), Endangered (EN - e.g., Bengal Tiger, Asian Elephant), Vulnerable (VU - e.g., One-Horned Rhinoceros), and Near Threatened (NT).
- **Major Threats to Biodiversity:**
  - **Habitat Loss and Fragmentation:** The single leading cause of species extinction worldwide.
  - **Poaching of Wildlife:** Illegal hunting for body parts (rhino horn, tiger bones, elephant ivory).
  - **Man-Wildlife Conflict:** Habitat encroachment causing carnivores and elephants to clash with humans.
  - **Biological Invasions:** Aggressive alien invasive species outcompeting native species (e.g., Water Hyacinth/Eichhornia crassipes, Lantana camara, Parthenium hysterophorus).
  - **Biomagnification of Pesticides:** Synthetic persistent organochlorines like DDT resist degradation and bioaccumulate in adipose tissue. Concentration multiplies across trophic levels, leading to eggshell thinning and reproductive collapse in predatory and

aquatic birds.

- **In-Situ vs Ex-Situ Conservation Strategies:**
  - In-Situ Conservation (On-Site): Preserves wild species within their natural native evolutionary habitats.
    - National Parks: Strictly protected areas managed by Central/State governments; no human settlements, grazing, or private timber extraction permitted (e.g., Jim Corbett NP in Uttarakhand - India's oldest, founded 1936; Kaziranga NP in Assam - famous for One-Horned Rhinos; Gir NP in Gujarat - Asiatic Lions).
    - Wildlife Sanctuaries: Protected areas focused on conserving particular species; limited traditional human activities like regulated cattle grazing and collection of minor forest produce may be permitted.
    - Biosphere Reserves: Large multi-purpose zones encompassing Core, Buffer, and Transition zones (e.g., Sundarbans in West Bengal, Nilgiri Biosphere Reserve).
    - Sacred Groves: Forest patches protected through indigenous religious traditions (e.g., Deorais in Maharashtra, Orans in Rajasthan).
  - Ex-Situ Conservation (Off-Site): Preserves threatened biological germplasm outside natural habitats in human-managed facilities:
    - Botanical Gardens, Zoological Parks, Aquariums, Seed Banks, Tissue Culture Laboratories, and Cryopreservation (storing gametes and embryos in liquid nitrogen at  $-196^{\circ}\text{C}$ ).
- **Key International Treaties & Wildlife Projects:**
  - Wildlife Protection Act (1972): India's principal statutory framework for wildlife conservation.
  - Project Tiger (1973): Initiated at Corbett NP to protect the royal Bengal tiger; expanded to 54+ reserves.
  - Ramsar Convention (1971): Global treaty dedicated to preserving international wetlands (e.g., Sundarbans, Chilika Lake, Keoladeo Ghana).

## Unit 5: Environmental Pollution & Waste Management — Core High-Yield Topic

**Exam Weightage: 42% of Exam | Approx 20-25 Questions (40-50 Marks)**

- **Air Pollution: Sources, Pollutants & Control:**
  - Primary Pollutants: Emitted directly from source into the air: Sulphur dioxide ( $\text{SO}_2$ ), Carbon monoxide ( $\text{CO}$ ), Nitrogen oxides ( $\text{NO}_x$ ), Hydrocarbons, and Particulate Matter (Fly Ash, soot).
  - Secondary Pollutants: Formed in ambient air through chemical or photochemical reactions among primary pollutants: Ground-level Ozone ( $\text{O}_3$ ), Peroxyacetyl Nitrate (PAN), Photochemical Smog, and Sulphuric Acid mists ( $\text{H}_2\text{SO}_4$ ).
  - Acid Rain:  $\text{SO}_2$  and  $\text{NO}_x$  emissions undergo oxidation in atmospheric moisture, forming dilute  $\text{H}_2\text{SO}_4$  and  $\text{HNO}_3$ . Rainfall with  $\text{pH} < 5.6$  is Acid Rain. Effects: Leaches soil nutrients, acidifies lakes killing aquatic life, destroys leaf chlorophyll, and chemically corrodes marble monuments ( $\text{CaCO}_3$  converted to soluble gypsum  $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ , called 'Stone Cancer', affecting the Taj Mahal).
  - Photochemical Smog: Brown haze formed on sunny, dry days via UV photolysis of vehicular  $\text{NO}_x$  and volatile organic compounds (VOCs), producing elevated levels of Ozone, PAN, aldehydes, and acrolein (causes acute eye irritation and respiratory distress).
  - Particulate Pollutants: Fly ash from coal-fired power plants. Control devices: Electrostatic Precipitators (ESPs - remove  $>99\%$  fine particles using high-voltage corona discharge), Cyclone Separators, Fabric Baghouse Filters, and Wet Venturi Scrubbers.
  - Gas Absorption: Activated Carbon beds utilize huge surface adsorption to capture odour compounds, VOCs, and chlorine.
- **Water Pollution: Parameters, Chemistry & Eutrophication:**
  - Point vs Non-Point Sources: Point sources discharge via discrete pipes (industrial effluent outfalls, municipal sewage plants). Non-point sources are diffuse landscape runoffs (agricultural fertilizer/pesticide runoff, urban street storm drainage).
  - Dissolved Oxygen (DO): Vital for aquatic fauna; healthy freshwater contains 7-10 mg/L DO at  $20^{\circ}\text{C}$ . If DO drops below 4.0 mg/L, water is deemed heavily polluted; fish mortality accelerates below 2.0 mg/L.
  - Biochemical Oxygen Demand (BOD): Milligrams of dissolved oxygen consumed by aerobic microorganisms to biologically oxidize organic waste in 1 litre of water during 5 days of incubation in dark at  $20^{\circ}\text{C}$  ( $\text{BOD}_5$ ). Clean drinking water:  $\text{BOD} < 1 \text{ mg/L}$ ; Treated sewage:  $\text{BOD} < 30 \text{ mg/L}$ ; Raw domestic sewage:  $\text{BOD} 200\text{-}400 \text{ mg/L}$ .
  - Chemical Oxygen Demand (COD): Milligrams of oxygen consumed to chemically oxidize ALL organic matter (both biologically degradable and non-biodegradable) using a strong chemical oxidant (acidified Potassium Dichromate,  $\text{K}_2\text{Cr}_2\text{O}_7$ ). For any water sample, COD is ALWAYS greater than or equal to BOD.
  - Eutrophication: Excessive nutrient enrichment of water bodies by Nitrates and Phosphates (from synthetic farm fertilizers and domestic phosphate detergents). Triggers explosive Algal Blooms that blanket the water surface, blocking sunlight. When algae die, decomposers consume all DO, suffocating fish.
  - Limiting Nutrient: In freshwater lakes, PHOSPHORUS is the primary limiting nutrient controlling algal blooms; in marine systems, NITROGEN is limiting.
  - Forms of Nitrogen: Wastewater contains 4 distinct forms: 1) Organic-N (proteins/urea), 2) Ammonia-N ( $\text{NH}_3/\text{NH}_4^+$ ), 3) Nitrite-N ( $\text{NO}_2^-$ ), and 4) Nitrate-N ( $\text{NO}_3^-$ ). Total Kjeldahl Nitrogen (TKN) = Organic-N + Ammonia-N.
- **Wastewater Treatment Engineering & Coagulation:**
  - Preliminary Treatment: Coarse Bar Screens remove large debris; Grit Chambers remove sand/gravel; Wastewater Pumping Stations lift sewage to provide hydraulic head.
  - Primary Treatment: Physical Primary Sedimentation Tanks allow settleable organic suspended solids to drop under gravity as raw

primary sludge (removes ~60% suspended solids, ~35% BOD).

- **Secondary Biological Treatment: Activated Sludge Process (ASP)** mixes settled wastewater with an active, flocculent biological suspension of aerobic bacteria and protozoa in an aeration tank. Air is continuously pumped to drive microbial bio-oxidation of dissolved organics, followed by a Secondary Clarifier where microbial flocs settle.
- **Coagulation / Flocculation:** Alum  $[\text{Al}_2(\text{SO}_4)_3 \cdot 18\text{H}_2\text{O}]$  added to alkaline water precipitates insoluble gelatinous Aluminium Hydroxide  $[\text{Al}(\text{OH})_3]$  flocs that entrap fine colloidal turbidity and neutralize negative electrical charges.
- **Disinfection:** Destroys pathogenic microorganisms prior to discharge or reuse; standard methods include Chlorination (leaves beneficial residual protection in swimming pools) and Ultraviolet (UV) irradiation.
- **Water Hardness:** Temporary Hardness is caused by dissolved Bicarbonates of Calcium and Magnesium  $[\text{Ca}(\text{HCO}_3)_2, \text{Mg}(\text{HCO}_3)_2]$  and is removable by boiling. Permanent Hardness is caused by dissolved Sulphates and Chlorides of Calcium and Magnesium  $(\text{MgSO}_4, \text{CaCl}_2)$  and requires lime-soda or ion-exchange softening.
- **Toxic Metals & Waterborne Health Hazards:**
  - **Minamata Disease:** Caused by bioaccumulated Methylmercury (Hg) in fish (first in Japan, 1956); causes central nervous system ataxia, numbness, blindness, and death.
  - **Itai-Itai Disease:** Caused by Cadmium (Cd) poisoning via contaminated rice crops; causes severe osteomalacia, bone fractures, and kidney failure.
  - **Blackfoot Disease:** Caused by high Arsenic ( $\text{As} > 0.05 \text{ mg/L}$ ) in groundwater aquifers of West Bengal and Bangladesh; causes peripheral vascular gangrene.
  - **Fluorosis:** Caused by excess Fluoride ( $\text{F} > 1.5 \text{ mg/L}$ ) in drinking water; causes dental mottling and crippling skeletal calcification.
  - **Blue Baby Syndrome (Methaemoglobinemia):** Caused by excess Nitrates ( $\text{NO}_3 > 45 \text{ mg/L}$ ) in infant formula, which oxidizes blood hemoglobin to non-functional methemoglobin.
  - **Waterborne Diseases:** Direct transmission through contaminated drinking water: Cholera (*Vibrio cholerae*), Typhoid (*Salmonella typhi*), Bacillary Dysentery, Hepatitis A & E, Amoebiasis.
  - **Vector-Borne Diseases:** NOT transmitted by drinking water: Malaria (transmitted by female *Anopheles* mosquito), Dengue (transmitted by *Aedes aegypti*).
- **Noise Pollution & Auditory Physiology:**
  - **Definition:** Unwanted, unpleasant, or damaging sound causing physiological annoyance or auditory trauma.
  - **Acoustic Measurement:** Sound Pressure Level is measured in Decibels (dB) on a logarithmic scale. 0 dB is the standardized threshold of human hearing (20  $\mu\text{Pa}$ ). Normal conversation is ~60 dB; traffic noise is ~80 dB; pain threshold begins at 120-130 dB.
  - **Auditory Health Impacts:** Exposure above 85 dB over prolonged periods causes permanent sensorineural hearing loss. Temporary short-term exposure causes Temporary Threshold Shift (TTS) — reversible auditory fatigue. Non-auditory effects include sleep disruption, hypertension, and tachycardia.
  - **Legal Status:** Noise was officially added as an atmospheric air pollutant in India via the 1987 Amendment to the Air Act, 1981.
- **Solid & Hazardous Waste Management:**
  - **Waste Hierarchy:** 3 R's — Reduce, Reuse, Recycle.
  - **Composting:** Aerobic biological decomposition of biodegradable organic waste by bacteria and fungi, converting it into rich humus (soil conditioner).
  - **Sanitary Landfill:** Engineered disposal of solid waste on land with clay/geomembrane bottom liners, daily soil compaction, and leachate/landfill gas collection systems. Buried organic matter decomposes via anaerobic methanogenesis, producing landfill gas ( $\text{CH}_4 + \text{CO}_2$ ).
  - **Incineration:** Controlled high-temperature thermal oxidation ( $850^\circ\text{C}$ - $1200^\circ\text{C}$ ) of non-biodegradable and biomedical waste, achieving complete pathogen destruction and volume reduction of up to 90%.
  - **Biomedical Waste:** Classified into categories (Yellow, Red, White, Blue). Sharps (needles, scalpels) present extreme hazards for percutaneous transmission of bloodborne viruses (Hepatitis B, Hepatitis C, HIV).
  - **E-Waste:** Obsolete computers, mobile devices, and printed circuit boards; contains toxic lead, cadmium, mercury, and halogenated flame retardants alongside recoverable precious metals.
- **Industrial & Nuclear Disasters:**
  - **Bhopal Gas Tragedy:** December 2-3, 1984, in Bhopal, Madhya Pradesh. Over 40 tonnes of Methyl Isocyanate (MIC,  $\text{CH}_3\text{NCO}$ ) leaked from the Union Carbide India Limited pesticide plant due to runaway exothermic reaction with water, killing thousands immediately and blinding/disabling tens of thousands.
  - **Chernobyl Disaster:** April 26, 1986, at Chernobyl Nuclear Power Plant, Ukraine (former USSR). Uncontrolled power surge caused steam explosions, core meltdown, and release of radioactive Caesium-137 and Iodine-131 across Europe.

## Unit 6: Environmental Policies, Legislation, Climate Change & Ozone Science

**Exam Weightage: 17% of Exam | Approx 8-10 Questions (16-20 Marks)**

- **Atmospheric Greenhouse Effect & Global Warming:**
  - **Mechanism:** Earth receives shortwave solar radiation, warms up, and radiates longwave infrared (thermal) radiation back into space. Greenhouse gases (GHGs) absorb this outgoing infrared radiation and re-radiate it downward, keeping Earth's surface warm ( $\sim 15^\circ\text{C}$  instead of a freezing  $-18^\circ\text{C}$ ).
  - **Key Anthropogenic GHGs:** Carbon dioxide ( $\text{CO}_2$  - responsible for ~60% of enhanced greenhouse effect), Methane ( $\text{CH}_4$  - GWP

28x), Nitrous oxide (N<sub>2</sub>O - GWP 265x), and synthetic Chlorofluorocarbons (CFCs / HFCs - GWPs in thousands).

- Impacts of Global Warming: Alpine and polar ice-sheet melting, thermal expansion of oceans causing sea-level rise, sinking of low-lying islands, shifting agricultural climate belts, and heightened cyclonic storm frequencies.
- IPCC: The Intergovernmental Panel on Climate Change (established 1988 by WMO and UNEP) publishes authoritative global scientific assessment reports evaluating climate risks.

- **Stratospheric Ozone Depletion:**
  - Chapman Cycle: Ozone (O<sub>3</sub>) is continuously generated and photodissociated in the Stratosphere (15-35 km) by ultraviolet radiation:  $O_2 + UV-C \rightarrow O + O$ ;  $O + O_2 \rightarrow O_3$ .
  - Chemical Depletion Mechanism: Synthetic Chlorofluorocarbons (CFC-11, CFC-12) used in refrigeration and aerosol propellants do not break down in the troposphere. In the stratosphere, intense UV photolyzes them, liberating free Chlorine radicals (Cl<sup>\*</sup>). A single chlorine radical catalytically destroys over 100,000 ozone molecules via catalytic cycles ( $Cl^* + O_3 \rightarrow ClO^* + O_2$ ;  $ClO^* + O \rightarrow Cl^* + O_2$ ).
  - Dobson Unit (DU): Measures the total overhead vertical column abundance of ozone gas. 1 DU = 0.01 mm thickness of pure ozone at standard temperature and pressure (STP). Normal atmospheric column thickness is ~300 DU (3 mm). An 'Ozone Hole' is defined when column thickness drops below 220 DU, observed annually over Antarctica.
  - Ecological Consequences: Depleted ozone allows intense UV-B radiation (280-315 nm) to reach Earth, causing human skin cancers (melanoma), corneal cataracts, immune suppression, and damage to terrestrial crop plants and marine phytoplankton.
  - Montreal Protocol: Adopted in 1987 in Montreal, Canada. A landmark international treaty that successfully mandated the phased elimination of production and consumption of ozone-depleting substances worldwide.

- **Indian Environmental Statutory Framework:**
  - Wildlife Protection Act (1972): Establishes National Parks and Wildlife Sanctuaries; strictly prohibits hunting; categorizes species into protection schedules.
  - Water (Prevention & Control of Pollution) Act (1974): Established the Central Pollution Control Board (CPCB) and State Pollution Control Boards (SPCBs); empowers authorities to inspect factories, establish discharge standards, and penalize polluters.
  - Forest (Conservation) Act (1980, amended 1988): Prohibits State Governments from de-reserving reserved forests or diverting forest land for non-forest purposes without prior approval from the Central Government.
  - Air (Prevention & Control of Pollution) Act (1981, amended 1987): Establishes ambient air quality standards and industrial emission limits; 1987 amendment brought Noise pollution under the legal definition of air pollutants.
  - Environment (Protection) Act (EPA, 1986): India's comprehensive 'Umbrella Legislation', enacted under Article 253 in the wake of the 1984 Bhopal disaster. Authorizes the Central Government to set nationwide standards, close non-compliant industrial facilities, and mandate Environmental Impact Assessments (EIA).
  - Environmental Impact Assessment (EIA): Statutory planning tool made mandatory under the EPA (1986) in 1994; evaluates potential environmental and social impacts of proposed developmental projects prior to granting clearance.
  - National Green Tribunal (NGT) Act (2010): Established specialized fast-track judicial courts for effective and expeditious disposal of cases relating to environmental protection and conservation, with a statutory mandate to decide cases within 6 months.
- **International Summits & Conventions:**
  - Stockholm UN Conference (1972): Pioneered global environmental multilateralism; led to UNEP and World Environment Day (June 5).
  - Rio Earth Summit / UNCED (1992): Generated Agenda 21 (action plan for sustainable development), the Rio Declaration, the Convention on Biological Diversity (CBD), and the UNFCCC.
  - Kyoto Protocol (1997): Under UNFCCC; established legally binding GHG reduction commitments for developed (Annex I) nations and introduced market-based Clean Development Mechanisms (CDM) and carbon credits.
  - Paris Agreement (2015, COP 21): Commits nations to hold global average temperature increase to well below 2.0°C above pre-industrial levels, pursuing efforts to limit it to 1.5°C.

## Unit 7: Human Communities, Environmental Movements & Disasters

**Exam Weightage: 3% of Exam | Approx 1-2 Questions (2-4 Marks)**

- **Human Population Growth & Demographics:** Global human population has exceeded 8 billion, driving massive resource consumption, municipal waste surges, and urban habitat encroachment beyond natural carrying capacities.
- **Resettlement & Rehabilitation (R&R) of Project-Affected Persons:** Large-scale infrastructural interventions (mega-dams, open-cast mines, industrial corridors) displace indigenous rural and tribal populations. Inadequate compensation, loss of cultural roots, and fractured livelihoods have driven major socio-environmental conflicts across India.
- **Disaster Management: Earthquakes & Floods:**
  - Floods: The most frequent, recurring, and widespread natural disaster in India and globally, accounting for over 40% of all disaster events (notably along the Ganga-Brahmaputra floodplains).
  - Earthquakes: Sudden release of strain energy along tectonic faults. Magnitude (total energy released) is measured on the logarithmic Richter Scale. Intensity (observed surface damage) is measured on the Modified Mercalli Intensity (MMI) Scale.
- **Historic Environmental Movements in India:**
  - Bishnoi Movement (1730, Khejarli, Rajasthan): Led by Amrita Devi Bishnoi, who along with 363 Bishnois sacrificed their lives hugging sacred Khejri trees to prevent the Maharaja of Jodhpur's men from felling

them.

- Chipko Movement (1973, Chamoli, Uttarakhand): Grassroots Gandhian non-violent resistance where rural Garhwali women hugged forest trees to stop commercial lumber contractors. Key figures: Sunderlal Bahuguna ('Ecology is permanent economy'), Chandi Prasad Bhatt, and Gaura Devi.
- Silent Valley Movement (1978, Kerala): Mobilization led by the Kerala Sastra Sahitya Parishad (KSSP) to stop a hydroelectric dam on the Kunthipuzha river, successfully preserving an evergreen tropical rainforest home to the endangered Lion-tailed Macaque.
- Narmada Bachao Andolan (NBA, 1985 onwards): Protest movement led by Medha Patkar and Baba Amte opposing the Sardar Sarovar Dam over the submergence of indigenous tribal lands and lack of rehabilitation.
- Appiko Movement (1983, Karnataka): Forest-hugging movement in the Western Ghats led by Panduranga Hegde, inspired directly by the Chipko movement.

- **Pioneers & Landmark Literature:** • Dr. M. S. Swaminathan: Father of the Green Revolution in India; introduced semi-dwarf high-yielding varieties of wheat and rice, transforming India from a food-deficit country to self-sufficiency.
- Dr. Salim Ali: The 'Birdman of India'; legendary ornithologist and author of 'The Fall of a Sparrow'.
- Rachel Carson: Author of 'Silent Spring' (1962), which documented synthetic pesticide biomagnification and launched the modern global environmental movement.
- Dr. Vandana Shiva: Prominent Indian ecofeminist and author of 'Staying Alive: Women, Ecology and Survival in India' (1988).
- **Constitutional Provisions for Environmental Protection in India:** • Article 48A (Directive Principles): 'The State shall endeavour to protect and improve the environment and to safeguard the forests and wild life of the country' (added by 42nd Amendment, 1976).
- Article 51A(g) (Fundamental Duties): 'It shall be the duty of every citizen of India to protect and improve the natural environment including forests, lakes, rivers and wild life, and to have compassion for living creatures'.
- Article 21 (Right to Life): The Supreme Court of India has interpreted the fundamental right to life to include the right to a clean, wholesome, pollution-free environment.

## SECTION 4: FULLY SOLVED 2024 EXAMINATION PAPER (SET D)

**Paper:** Bachelor of Engineering Examination, 2024 (1st Year, 2nd Semester, Compulsory EVS, Code: D)

**Full Marks:** 100 | **Questions:** 50 MCQs (2 Marks each) | **Negative Marking:** NIL

**Q1. Sulphur dioxide and nitrogen oxides are transported by prevailing winds; they form secondary pollutants such as nitric acid vapour and droplets of sulfuric acid descending on earth surface. This phenomenon is called:**

- (a) Rain
- (b) Acid rain
- (c) Basic rain
- (d) Ice rain

**Correct Answer: Option (b)**

**Explanation:** Acid rain occurs when  $\text{SO}_2$  and  $\text{NO}_x$  react with atmospheric water vapor, forming  $\text{H}_2\text{SO}_4$  and  $\text{HNO}_3$ . Rainfall with  $\text{pH} < 5.6$  is defined as acid rain. It damages aquatic systems and corrodes marble monuments (Stone Cancer).

**Q2. Ozone layer is present in which atmospheric layer?**

- (a) Troposphere
- (b) Mesosphere
- (c) Lithosphere
- (d) Stratosphere

**Correct Answer: Option (d)**

**Explanation:** The ozone layer resides primarily in the Stratosphere (15 to 35 km altitude). It absorbs lethal Solar UV-B and UV-C radiation. Tropospheric ozone, by contrast, is a harmful secondary air pollutant and greenhouse gas.

**Q3. WWF stands for:**

- (a) World Wide Fund
- (b) World Wildlife Fund for Nature
- (c) Wildlife Wide Fund for Nature and Natural Resources
- (d) World Wild Fund

**Correct Answer: Option (b)**

**Explanation:** WWF stands for World Wildlife Fund for Nature (founded 1961 in Switzerland). Its famous global emblem is the Giant Panda.

**Q4. Which National Park is famous for having the Great Indian One-Horned Rhinoceros?**

- (a) Rajaji National Park
- (b) Jim Corbett National Park
- (c) Bandipur National Park
- (d) Kaziranga National Park

**Correct Answer: Option (d)**

**Explanation:** Kaziranga National Park in Assam hosts over two-thirds of the world's Great One-Horned Rhinoceroses (*Rhinoceros unicornis*) and is a designated UNESCO World Heritage Site.

**Q5. Endemic species are defined as:**

- (a) Rare species
- (b) Cosmopolitan distribution
- (c) Species localized in a specific geographic region
- (d) Critically endangered species

**Correct Answer: Option (c)**

**Explanation:** Endemic species are plants or animals restricted exclusively to a defined geographic area (e.g., Lion-tailed macaque in the Western Ghats) and found nowhere else in the wild.

**Q6. Which of the following are classified as autotrophs?**

- (a) Green plants
- (b) Animals
- (c) Birds
- (d) All the above

**Correct Answer: Option (a)**

**Explanation:** Autotrophs (primary producers) synthesize their own organic food from inorganic substances ( $\text{CO}_2$ ,  $\text{H}_2\text{O}$ ) using solar energy through photosynthesis. Green plants and cyanobacteria are autotrophs.

**Q7. \_\_\_\_\_ is the second generation of synthetic pesticides which enters and persists in the food chain:**

- (a) Malathion
- (b) Boric Acid
- (c) Diazinon
- (d) DDT

**Correct Answer: Option (d)**

**Explanation:** DDT (Dichlorodiphenyltrichloroethane) is a synthetic organochlorine insecticide. It resists biological degradation, bioaccumulates in fatty tissues, and biomagnifies up trophic levels.

**Q8. IUCN stands for:**

- (a) International Union for Conservation of Nature
- (b) International Union for Conservation of Nature and Natural Resources
- (c) International Union Council for Nature
- (d) International Union Council for Conservation of Nature

**Correct Answer: Option (b)**

**Explanation:** IUCN officially stands for International Union for Conservation of Nature and Natural Resources (headquartered in Gland, Switzerland). It compiles and publishes the global Red List.

**Q9. Society should become educated and aware of the environmental dangers of:**

- (a) Population explosion
- (b) Deforestation and ozone layer depletion
- (c) Industrialization and global warming
- (d) All the above

**Correct Answer: Option (d)**

**Explanation:** Environmental education requires holistic awareness of anthropogenic drivers including explosive population growth, resource depletion, industrial emissions, and planetary warming.

**Q10. In which year was the landmark U.N. Conference on Human Environment held at Stockholm?**

- (a) 1962
- (b) 1972
- (c) 1982
- (d) 1992

**Correct Answer: Option (b)**

**Explanation:** The Stockholm Conference took place in June 1972. It was the first global summit on international environmental issues, creating the United Nations Environment Programme (UNEP).

**Q11. Which specialized anatomical part of a plant transpires/evaporates water into the atmosphere?**

- (a) Stomata
- (b) Fruit
- (c) Branch
- (d) Root

**Correct Answer: Option (a)**

**Explanation:** Stomata are microscopic pores located primarily on plant leaf epidermis that regulate gas exchange (CO<sub>2</sub> and O<sub>2</sub>) and drive transpiration pull, releasing water vapor.

**Q12. Which primary natural meteorological factor causes drought?**

- (a) Over-cultivation
- (b) Rain failure
- (c) Overgrazing
- (d) None of the above

**Correct Answer: Option (b)**

**Explanation:** Rain failure (monsoon deficit or prolonged precipitation absence) is the primary natural cause of meteorological drought. Overgrazing and over-cultivation are anthropogenic accelerants.

**Q13. Which major Indian dam is situated in a geographic zone of high seismicity?**

- (a) Sardar Sarovar Dam
- (b) Nagarjuna Sagar Dam
- (c) Bhakra Nangal Dam
- (d) Tehri Dam

**Correct Answer: Option (d)**

**Explanation:** Tehri Dam (built on the Bhagirathi River in Uttarakhand) is situated in Seismic Zone V of the central Himalayan seismic gap, creating serious seismic hazard concerns.

**Q14. Disposal of any kind of discarded electrical and electronic equipment is designated as:**

- (a) General Waste
- (b) Domestic Waste
- (c) Vegetable Waste
- (d) e-waste

**Correct Answer: Option (d)**

**Explanation:** e-waste (electronic waste) encompasses discarded electronic devices (computers, mobile phones, printed circuit boards). It contains toxic heavy metals (lead, cadmium, mercury).

**Q15. The principal hydrological objective of rainwater harvesting is to recharge the:**

- (a) Public gardens
- (b) Surface wells
- (c) Irrigation canals
- (d) Groundwater table

**Correct Answer: Option (d)**

**Explanation:** Rainwater harvesting captures surface runoff to artificially recharge sub-surface aquifers, counteracting rapid water table depletion caused by borewell over-extraction.

**Q16. Jim Corbett National Park was established in which Indian state?**

- (a) Uttar Pradesh
- (b) Tamil Nadu
- (c) Gujarat
- (d) Uttarakhand

**Correct Answer: Option (d)**

**Explanation:** Jim Corbett National Park was established in 1936 (originally Hailey National Park) in the Nainital and Pauri Garhwal districts of Uttarakhand. It was the launch park for Project Tiger (1973).

**Q17. The controlled high-temperature thermal destruction of solid waste is called:**

- (a) Composting
- (b) Incineration
- (c) Segregation
- (d) Sanitary land filling

**Correct Answer: Option (b)**

**Explanation:** Incineration is the controlled combustion of solid waste at 850°C–1200°C. It is mandatory for biomedical waste because it ensures complete sterilization and destruction of pathogenic organisms.

**Q18. Which clean energy source is widely recognized as the future transport fuel for mankind?**

- (a) Biofuels
- (b) Hydrogen fuels
- (c) Biogas
- (d) Biomass energy

**Correct Answer: Option (b)**

**Explanation:** Hydrogen fuel ( $H_2$ ) possesses high gravimetric energy density and emits only water vapor ( $H_2O$ ) upon combustion, making green hydrogen the ultimate clean future energy carrier.

**Q19. Deforestation directly leads to a significant regional reduction of:**

- (a) Rainfall
- (b) Soil erosion
- (c) Groundwater depletion
- (d) All the above

**Correct Answer: Option (a)**

**Explanation:** Forests cycle vast quantities of moisture into the atmosphere via evapotranspiration. Large-scale deforestation disrupts this biotic pump, leading to decreased regional precipitation and rainfall.

**Q20. The ecological and societal functions of forest ecosystems include:**

- (a) Productive functions
- (b) Protective functions
- (c) Regulative functions
- (d) All the above

**Correct Answer: Option (d)**

**Explanation:** Forests provide timber/fruits/resin (productive), soil and watershed conservation (protective), and carbon sequestration/climate and hydrological stabilization (regulative).

**Q21. Which of the following is classified as a renewable, yet exhaustible natural resource?**

- (a) Forest
- (b) Coal
- (c) Petroleum
- (d) Minerals

**Correct Answer: Option (a)**

**Explanation:** Forests and biological resources regenerate naturally via reproduction (renewable), but can be completely exhausted if harvesting rates exceed their natural regeneration capacity.

**Q22. The standard and most suitable method for the disinfection of public swimming pool water is:**

- (a) Ultra-violet rays treatment
- (b) Lime treatment
- (c) Potassium permanganate
- (d) Chlorination

**Correct Answer: Option (d)**

**Explanation:** Chlorination (using chlorine gas, sodium hypochlorite, or chloramines) provides a vital residual disinfectant concentration in water, preventing continuous re-contamination by swimmers.

**Q23. Hydropower generation plants convert the kinetic and potential energy of falling water into:**

- (a) Mechanical energy
- (b) Tidal energy
- (c) Wind energy
- (d) Electricity

**Correct Answer: Option (d)**

**Explanation:** Water head drives hydraulic turbines (converting kinetic/potential energy to mechanical rotation), which drive electromagnetic generators to produce electrical power.

**Q24. Which of the following 3 R's represents the internationally recognized waste hierarchy?**

- (a) Reduce, Rebuild, Restrict
- (b) Reduce, Reuse, Recycle
- (c) Read, Register, Recall
- (d) Random, Reduce, Recall

**Correct Answer: Option (b)**

**Explanation:** The classical waste minimization hierarchy prioritizes: 1) Source Reduction, 2) Direct Reusing of materials, and 3) Material Recycling into new commodities.

**Q25. Which toxic air pollutant gas destroys chlorophyll and causes severe chlorosis in plant leaves?**

- (a) CO
- (b) CO<sub>2</sub>
- (c) SO<sub>2</sub>
- (d) H<sub>2</sub>S

**Correct Answer: Option (c)**

**Explanation:** Sulphur dioxide (SO<sub>2</sub>) penetrates leaf stomata and converts to bisulphite and sulphite ions, bleaching chlorophyll pigments and causing leaf necrosis and chlorosis.

**Q26. Anthropogenic global warming is predicted to cause which of the following ecological crises?**

- (a) Melting of polar glaciers
- (b) Rising mean sea levels
- (c) Submergence of low-lying islands
- (d) All the above

**Correct Answer: Option (d)**

**Explanation:** Thermal expansion of oceans combined with continental ice-sheet melting raises sea levels, threatening low-lying coastal cities and island nations (e.g., Maldives, Tuvalu).

**Q27. The municipal water purification step known as 'Disinfection' specifically achieves:**

- (a) Complete removal of turbidity
- (b) Removal of mineral hardness
- (c) Killing of pathogenic disease-causing bacteria
- (d) Complete chemical sterilization

**Correct Answer: Option (c)**

**Explanation:** Disinfection specifically destroys or inactivates disease-causing pathogenic bacteria, viruses, and amoebic cysts. It differs from absolute sterilization, which eliminates all living matter.

**Q28. The most common natural dissolved gas responsible for natural acidity in rainwater and surface water is:**

- (a) Carbon dioxide
- (b) Oxygen
- (c) Hydrogen
- (d) Nitrogen

**Correct Answer: Option (a)**

**Explanation:** Atmospheric CO<sub>2</sub> dissolves in water to establish chemical equilibrium:  $\text{CO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{H}_2\text{CO}_3$  (carbonic acid), yielding a natural unpolluted rainwater pH of approximately 5.6.

**Q29. Which of the following infectious human diseases is NOT a water-borne disease?**

- (a) Dysentery
- (b) Cholera
- (c) Typhoid
- (d) Malaria

**Correct Answer: Option (d)**

**Explanation:** Malaria is a vector-borne protozoan infection transmitted through the bite of infected female *Anopheles* mosquitoes. Cholera, typhoid, and dysentery are direct waterborne bacterial infections.

**Q30. Environmentally, 'polluted water' is defined as water that:**

- (a) Contains only harmless inert silt
- (b) Contains undesirable substances rendering it unfit for domestic or drinking use
- (c) Is safe and suitable for drinking
- (d) Is purely distilled

**Correct Answer: Option (b)**

**Explanation:** Water pollution is defined as any physical, chemical, or biological alteration in water quality that produces harmful effects on living organisms or renders water unfit for beneficial consumption.

**Q31. Who is universally celebrated as the 'Father of the Green Revolution in India'?**

- (a) Dr. M. S. Swaminathan
- (b) Harikrishna Jain
- (c) Vandana Shiva
- (d) Dr. B. P. Pal

**Correct Answer: Option (a)**

**Explanation:** Dr. M. S. Swaminathan introduced high-yielding varieties (HYVs) of semi-dwarf wheat and rice (collaborating with Norman Borlaug) in the 1960s, making India food self-sufficient.

**Q32. Which non-edible plant seed oil is widely cultivated and processed in India as biodiesel?**

- (a) Castor oil
- (b) Jatropha oil
- (c) Cottonseed oil
- (d) Flaxseed oil

**Correct Answer: Option (b)**

**Explanation:** *Jatropha curcas* is a hardy drought-resistant shrub whose seeds yield ~35% non-edible vegetable oil. Transesterification of *Jatropha* oil produces commercial biodiesel.

**Q33. Stagnant water forming breeding grounds for \_\_\_\_\_ mosquitoes is the vector for malaria:**

- (a) Aedes
- (b) Anopheles
- (c) Culex
- (d) Sabethes

**Correct Answer: Option (b)**

**Explanation:** Female *Anopheles* mosquitoes breed in clean, stagnant pools and transmit *Plasmodium* parasites. *Aedes* transmits Dengue and Chikungunya; *Culex* transmits Filariasis.

**Q34. The landmark Water (Prevention and Control of Pollution) Act was enacted by Indian Parliament in:**

- (a) 1974
- (b) 1975
- (c) 1986
- (d) 1994

**Correct Answer: Option (a)**

**Explanation:** Enacted in 1974, this foundational statute established the institutional machinery of the Central Pollution Control Board (CPCB) and State Pollution Control Boards (SPCBs).

**Q35. The most destructive synthetic chemical class responsible for stratospheric ozone depletion is:**

- (a) CFCs (Chlorofluorocarbons)
- (b) CH<sub>4</sub>
- (c) N<sub>2</sub>O
- (d) All the above

**Correct Answer: Option (a)**

**Explanation:** CFCs diffuse unchanged to the stratosphere where UV radiation photolyzes them, releasing free Chlorine radicals (Cl<sup>\*</sup>). One Chlorine radical catalytically destroys over 100,000 ozone molecules.

**Q36. The international scientific body abbreviation IPCC stands for:**

- (a) Information Processing Command and Control
- (b) Interprocess Communication and Control
- (c) Intergovernmental Panel on Climate Change
- (d) Integrated Panel on Climate Change

**Correct Answer: Option (c)**

**Explanation:** Established in 1988 by the WMO and UNEP, the IPCC provides rigorous scientific assessments on anthropogenic climate change, greenhouse emissions, and mitigation strategies.

**Q37. Which lethal toxic gas leaked from the Union Carbide pesticide plant in the Bhopal Gas Tragedy?**

- (a) Hydrogen fluoride
- (b) Methyl chloride
- (c) Hydrogen cyanide
- (d) Methyl Isocyanate (MIC)

**Correct Answer: Option (d)**

**Explanation:** On the night of Dec 2-3, 1984, over 40 metric tons of volatile Methyl Isocyanate (MIC,  $\text{CH}_3\text{NCO}$ ) gas leaked in Bhopal, causing the deadliest industrial disaster in human history.

**Q38. How do buried biodegradable organic solid wastes decompose inside a sanitary landfill?**

- (a) By the biochemical action of microorganisms
- (b) By physical friction with soil particles
- (c) By the flow of groundwater
- (d) By atmospheric direct oxidation

**Correct Answer: Option (a)**

**Explanation:** Buried organic matter undergoes initial aerobic digestion followed by prolonged anaerobic microbial decomposition (methanogenesis), generating landfill gas ( $\text{CH}_4 + \text{CO}_2$ ).

**Q39. Which of the following post-consumer materials can be re-melted and recycled indefinitely?**

- (a) Organic materials
- (b) Aluminum
- (c) Wood
- (d) Synthetic plastics

**Correct Answer: Option (b)**

**Explanation:** Aluminum can be melted and recycled repeatedly with zero degradation of structural properties, saving up to 95% of the energy required for virgin smelting from bauxite ore.

**Q40. The eggs of raptors and aquatic piscivorous birds fail to hatch primarily due to contamination by:**

- (a) Malathion
- (b)  $\text{SO}_2$
- (c) DDT
- (d) Potassium chloride

**Correct Answer: Option (c)**

**Explanation:** DDT biomagnifies in aquatic food webs. In top predatory birds (eagles, pelicans), DDE disrupts calcium metabolism and carbonic anhydrase, causing eggshell thinning and crushing.

**Q41. The critical fundamental water quality parameter required for sustaining higher aquatic life is:**

- (a) Dissolved Oxygen (DO)
- (b) Dissolved Chlorine
- (c) Dissolved Nitrogen
- (d) Dissolved Methane

**Correct Answer: Option (a)**

**Explanation:** Fish and aquatic organisms require a minimum Dissolved Oxygen concentration of 4.0 to 6.0 mg/L for respiration. Depletion below 2.0 mg/L causes hypoxia and massive fish kills.

**Q42. Urban and vehicular air pollution can be mitigated through:**

- (a) Modification of industrial emission control equipment
- (b) Using eco-friendly Biodiesel and clean fuels
- (c) Installation of catalytic converters and particulate traps
- (d) All the above

**Correct Answer: Option (d)**

**Explanation:** Effective air pollution abatement requires holistic control: scrubbers/ESPs in factories, cleaner fuel alternatives (CNG/biodiesel), and automotive catalytic exhaust treatments.

**Q43. Which of the following is classified as a primary air pollutant?**

- (a) Peroxyacetyl Nitrate (PAN)
- (b) Photochemical Smog
- (c) Ground-level Ozone
- (d) Sulphur oxides (SO<sub>x</sub>)

**Correct Answer: Option (d)**

**Explanation:** Primary pollutants (SO<sub>x</sub>, NO<sub>x</sub>, CO) are emitted directly from identifiable point sources. PAN, ozone, and photochemical smog are secondary pollutants formed via atmospheric reactions.

**Q44. June 5th is observed internationally across the globe as:**

- (a) World Environment Day
- (b) World Forest Day
- (c) World Wildlife Day
- (d) World Population Day

**Correct Answer: Option (a)**

**Explanation:** Established by the United Nations General Assembly in 1972 on the opening day of the Stockholm Conference, June 5 is the premier platform for global environmental outreach.

**Q45. In Indian environmental regulatory administration, CPCB stands for:**

- (a) Committee of Pollution Control Board
- (b) Central Pollution Control Board
- (c) Central Population Control Board
- (d) Committee for Protection of Conservation Bodies

**Correct Answer: Option (b)**

**Explanation:** The Central Pollution Control Board (CPCB) is India's apex statutory regulatory body for environmental monitoring, air and water standards enforcement, established under the Water Act 1974.

**Q46. Accelerated polar temperature rise and ice-cap melting is primarily driven by:**

- (a) Global warming
- (b) Stratospheric ozone depletion
- (c) Acid precipitation
- (d) Carbon monoxide build-up

**Correct Answer: Option (a)**

**Explanation:** Greenhouse gas-induced global warming causes polar amplification (ice-albedo positive feedback), accelerating Arctic and Antarctic ice loss and sea-level rise.

**Q47. Which of the following groups comprises the principal Greenhouse Gases (GHGs)?**

- (a) CO, CO<sub>2</sub>, Methane and Nitrogen
- (b) CO, SO<sub>2</sub>, Nitric oxide and Methane
- (c) Carbon dioxide, Nitrous oxide, Methane and CFCs
- (d) CO<sub>2</sub>, SO<sub>2</sub>, CFCs and Argon

**Correct Answer: Option (c)**

**Explanation:** The Kyoto Protocol identifies 6 primary GHGs: Carbon dioxide (CO<sub>2</sub>), Methane (CH<sub>4</sub>), Nitrous oxide (N<sub>2</sub>O), Hydrofluorocarbons (HFCs), Perfluorocarbons (PFCs), and Sulphur hexafluoride (SF<sub>6</sub>).

**Q48. Which international conservation organization compiles and publishes the Red Data Book?**

- (a) IUCN
- (b) NEERI
- (c) NWAP
- (d) CITES

**Correct Answer: Option (a)**

**Explanation:** The IUCN (International Union for Conservation of Nature) Red List of Threatened Species was established in 1964 and remains the authoritative inventory of global species extinction risk.

**Q49. In municipal and industrial water treatment, Activated Carbon is utilized primarily for:**

- (a) Destruction of pathogenic bacteria
- (b) Removing temporary hardness
- (c) Adsorption of odours, tastes and synthetic organics
- (d) Preventing pipe corrosiveness

**Correct Answer: Option (c)**

**Explanation:** Activated carbon boasts an extraordinarily high internal surface area (500-1500 m<sup>2</sup>/g), allowing powerful physical adsorption of volatile organics, chlorine, odour-producing compounds, and micropollutants.

**Q50. The total vertical column abundance of atmospheric ozone is measured in units of:**

- (a) Angstrom Units
- (b) Light Years
- (c) Dobson Units (DU)
- (d) Archimedes Units

**Correct Answer: Option (c)**

**Explanation:** One Dobson Unit (DU) corresponds to a 0.01 mm thickness of pure ozone gas compressed to standard temperature and pressure (STP). Normal atmospheric ozone column is 300 DU (~3 mm thick).

## SECTION 5: FULLY SOLVED 2023 EXAMINATION PAPER (SETS C & D)

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**Paper:** Bachelor of Arts Examination, 2023 (1st Year, 1st Semester, Compulsory EVS, Sets C & D)

**Full Marks:** 100 | **Questions:** 50 MCQs (2 Marks each) | **Negative Marking:** NIL

*Note: Sets C and D feature the exact same 50 questions jumbled in different sequential orders.*

**Q1. Jim Corbett National Park was established in which Indian state?**

- (a) Uttar Pradesh
- (b) Tamil Nadu
- (c) Gujarat
- (d) Uttarakhand

**Correct Answer: Option (d)**

**Explanation:** Jim Corbett National Park was founded in 1936 (as Hailey NP) in Uttarakhand. It is India's first national park and the birthplace of Project Tiger.

**Q2. The most common traditional domestic energy source in rural Indian villages is:**

- (a) Coal
- (b) Biogas
- (c) Electricity
- (d) Fuelwood and animal dung

**Correct Answer: Option (d)**

**Explanation:** Over 60-70% of rural Indian households traditionally depend on biomass fuels—primarily gathered wood and dried cattle dung cakes—for domestic cooking.

**Q3. The statutory provisions of the Wildlife Protection Act (1972) encompass:**

- (a) Prohibition of hunting and commercial exploitation of wild animals
- (b) Constitution of National Parks, Sanctuaries, and Conservation Reserves
- (c) Special legal schemes for preserving endangered species
- (d) All the above

**Correct Answer: Option (d)**

**Explanation:** The Act provides comprehensive protection: prohibition of hunting (Section 9), creation of protected areas (Chapter IV), and penal schedules for offenses.

**Q4. The Indian Wildlife Protection Act was officially enacted in which year?**

- (a) 1952
- (b) 1972
- (c) 1980
- (d) 1970

**Correct Answer: Option (b)**

**Explanation:** Passed by the Indian Parliament in 1972 to safeguard wildlife habitats, regulate trade in wildlife products, and enforce penal action against poaching.

**Q5. All living biological organisms present within an environment are categorized as:**

- (a) Abiotic elements
- (b) Biotic elements
- (c) Microclimatic factors
- (d) Edaphic elements

**Correct Answer: Option (b)**

**Explanation:** Biotic elements comprise all living components (producers, consumers, decomposers). Abiotic elements are non-living physical/chemical factors (sunlight, water, air, soil).

**Q6. The annual Indian conservation movement 'Van Mahotsav' encompasses:**

- (a) Planting and actively protecting native trees
- (b) Curtailing wanton forest destruction
- (c) Restoring ecological green cover across landscapes
- (d) All the above

**Correct Answer: Option (d)**

**Explanation:** Van Mahotsav (Festival of Trees) was initiated in 1950 by Union Minister K.M. Munshi. It is celebrated across India during the first week of July.

**Q7. The historic Chipko Movement originated in the \_\_\_\_\_ district of the Garhwal Himalayas:**

- (a) Haridwar
- (b) Dehradun
- (c) Champawat
- (d) Chamoli

**Correct Answer: Option (d)**

**Explanation:** The Chipko movement began in 1973 in Reni village, Chamoli district, Uttarakhand, where rural women led by Gaura Devi hugged forest trees to prevent commercial logging.

**Q8. Which of the following is a classic example of non-renewable natural resources?**

- (a) Surface water
- (b) Solar radiation
- (c) Agricultural crops
- (d) Coal and petroleum

**Correct Answer: Option (d)**

**Explanation:** Fossil fuels (coal, petroleum, natural gas) formed over hundreds of millions of years and cannot be replenished within a human timescale (finite non-renewable stock).

**Q9. Which toxic metalloid element present in alluvial groundwater causes widespread fatal diseases in the Ganga-Brahmaputra delta?**

- (a) Phosphorus
- (b) Calcium
- (c) Arsenic
- (d) Sodium

**Correct Answer: Option (c)**

**Explanation:** Arsenic ( $As > 0.05$  mg/L) dissolved from alluvial aquifer sediments causes chronic arsenicosis, skin lesions, blackfoot disease, and multiple internal cancers.

**Q10. In which year was the Forest (Conservation) Act substantially amended by Parliament?**

- (a) 1978
- (b) 1988
- (c) 1963
- (d) 1952

**Correct Answer: Option (b)**

**Explanation:** Originally enacted in 1980, the Forest Conservation Act was amended in 1988 to strictly prohibit leasing or diversion of forest land to non-governmental agencies without prior Central approval.

**Q11. Under the Air (Prevention and Control of Pollution) Act, which gaseous emissions are strictly regulated and monitored?**

- (a) Sulphur Dioxide ( $SO_2$ )
- (b) Carbon Monoxide (CO)
- (c) Nitrogen Oxides ( $NO_x$ )
- (d) All the above

**Correct Answer: Option (d)**

**Explanation:** The Air Act regulates industrial and vehicular emissions of criteria air pollutants:  $SO_2$ ,  $NO_x$ , CO, and particulate matter, setting statutory National Ambient Air Quality Standards.

**Q12. Which stimulant alkaloid is commercially extracted from tea (*Camellia sinensis*) or coffee (*Coffea arabica*) plants?**

- (a) Camphor
- (b) Caffeine
- (c) Thorn Apple (Atropine)
- (d) Morphine

**Correct Answer: Option (b)**

**Explanation:** Caffeine is a purine alkaloid that functions as a natural pesticide in tea and coffee plants, acting as a central nervous system stimulant in humans.

**Q13. In which geographic physiographic setting do we characteristically encounter brackish water ecosystems?**

- (a) Alpine wetlands
- (b) Fast mountain streams
- (c) River deltas and estuaries
- (d) Deep oceanic trenches

**Correct Answer: Option (c)**

**Explanation:** Brackish water ecosystems (salinity 0.5 to 30 ppt) form at river deltas, estuaries, and coastal lagoons where incoming freshwater mixes with oceanic saltwater.

**Q14. Progressive ecological degradation and desertification in the Cauvery basin is driven by:**

- (a) Deforestation in the Western Ghats catchments
- (b) Rapid urbanization and water extraction
- (c) Expanding intensive agriculture and industrial discharge
- (d) All the above

**Correct Answer: Option (d)**

**Explanation:** River catchment degradation results from compounding pressures: upstream forest felling, heavy sand mining, industrial pollution, and overallocation of irrigation waters.

**Q15. Which primary natural meteorological factor triggers agricultural and hydrological drought?**

- (a) Over-cultivation
- (b) Rain failure (monsoon deficit)
- (c) Overgrazing
- (d) None of the above

**Correct Answer: Option (b)**

**Explanation:** Rain failure or failure of the monsoon rains is the prime natural driver of drought. Land-use practices like over-cultivation exacerbate drought impacts.

**Q16. Which concrete gravity dam is historically cited as one of the largest multipurpose dams in India?**

- (a) Tehri Dam
- (b) Bhakra Nangal Dam
- (c) Sardar Sarovar Dam
- (d) Nagarjuna Sagar Dam

**Correct Answer: Option (b)**

**Explanation:** Bhakra Dam on the Sutlej River (Himachal Pradesh) is India's second tallest dam (226 m) and its largest concrete gravity dam. (Note: Tehri is the tallest rock-fill dam at 260 m).

**Q17. Which high-altitude dam is constructed in an active seismic fault zone of the Himalayas?**

- (a) Sardar Sarovar Dam
- (b) Nagarjuna Sagar Dam
- (c) Bhakra Nangal Dam
- (d) Tehri Dam

**Correct Answer: Option (d)**

**Explanation:** The 260-meter high Tehri rock-and-earth fill dam on the Bhagirathi River sits directly in Seismic Zone V, raising major concerns regarding potential earthquake-induced dam failure.

**Q18. What major socio-ecological challenge is created in the upstream catchment when mega dams are constructed?**

- (a) Indigenous tribal populations face forced displacement from ancestral lands
- (b) Extensive valleys of pristine flora and fauna are permanently submerged
- (c) Stagnant reservoir margins increase vector breeding and waterborne disease
- (d) All the above

**Correct Answer: Option (d)**

**Explanation:** Large reservoirs inundate millions of hectares of arable land and forests upstream, displacing indigenous populations (Rehabilitation and Resettlement crisis) and destroying habitats.

**Q19. \_\_\_\_\_ are considered the cultural and ecological lifelines and pride of India:**

- (a) Rivers
- (b) Oceans
- (c) Canals
- (d) Farm ponds

**Correct Answer: Option (a)**

**Explanation:** India's river systems (Ganga, Brahmaputra, Indus, Godavari, Krishna, Narmada, Cauvery) have sustained civilization, agriculture, and ecosystems for millennia.

**Q20. Which day is observed worldwide as World Environment Day?**

- (a) July 5th
- (b) June 10th
- (c) October 20th
- (d) June 5th

**Correct Answer: Option (d)**

**Explanation:** World Environment Day is celebrated every year on June 5th to commemorate the 1972 Stockholm Conference on the Human Environment.

**Q21. Which of the following does NOT constitute a causal reason for the loss of forests?**

- (a) Extinction of endangered species
- (b) Commercial timber harvesting and logging
- (c) Construction of mega dams and open-cast mining
- (d) Fuelwood extraction by rural populations

**Correct Answer: Option (a)**

**Explanation:** Extinction of species is a devastating \*consequence\* of forest habitat destruction, whereas mining, dams, logging, and fuelwood gathering are direct \*causes\* of deforestation.

**Q22. According to standard forestry classification, which of the following is an INDIRECT use of forest products?**

- (a) Grass harvested for grazing and stall feeding
- (b) Gums and resins processed into industrial commodities
- (c) Wild medicinal herbs gathered for healing
- (d) Bamboo cut for basketry

**Correct Answer: Option (b)**

**Explanation:** As classified in Erach Bharucha's UGC textbook (Page 96): Grazing fodder, bamboo, and medicinal herbs used directly by forest dwellers are direct uses; gums/resins processed into commercial items are indirect uses.

**Q23. Which primary energy source represents the largest share of commercial energy consumption worldwide?**

- (a) Nuclear energy
- (b) Natural gas
- (c) Oil (Petroleum)
- (d) Coal

**Correct Answer: Option (c)**

**Explanation:** Globally, petroleum oil remains the single largest commercial energy source (accounting for ~33% to 39%), followed by coal (~27%) and natural gas (~24%).

**Q24. An intricate network of interconnected and overlapping food chains in an ecosystem forms a:**

- (a) Food pyramid
- (b) Food web
- (c) Carbon cycle
- (d) Nutrient sink

**Correct Answer: Option (b)**

**Explanation:** In nature, feeding relationships are rarely isolated linear chains; multiple interconnected food chains form a stable, resilient ecological 'Food Web'.

**Q25. What is the targeted optimal forest cover that India should achieve under the National Forest Policy?**

- (a) 15%
- (b) 50%
- (c) 44%
- (d) 33%

**Correct Answer: Option (d)**

**Explanation:** The National Forest Policy of 1988 mandates that a minimum of 33% of India's total geographical area should be under forest or tree cover (67% in mountainous regions, 20% in plains).

**Q26. Approximately what percentage of the Earth's total planetary surface is covered by oceans and water bodies?**

- (a) 71%
- (b) 74%
- (c) 65%
- (d) 80%

**Correct Answer: Option (a)**

**Explanation:** Approximately 71% of Earth's surface (~361 million km<sup>2</sup>) is covered by water, of which 97.5% is saline ocean water and only 2.5% is freshwater.

**Q27. The extraction of unconventional fossil fuels (shale gas/tight oil) from deep rock formations employs:**

- (a) Hydroelectric generation
- (b) Hydraulic fracturing (Fracking)
- (c) Fractional distillation
- (d) Surface open-pit mining

**Correct Answer: Option (b)**

**Explanation:** Hydraulic fracturing injects high-pressure fluid (water, chemical additives, and sand) deep underground to fracture shale rock strata and release trapped hydrocarbons.

**Q28. Biogas generated via the anaerobic digestion of cattle dung and agro-residues is commonly called:**

- (a) Producer gas
- (b) Water gas
- (c) Gobar gas
- (d) Coal gas

**Correct Answer: Option (c)**

**Explanation:** In India, biogas plants utilizing bovine excreta (gobar) are universally termed Gobar Gas plants. The gas contains 55-70% methane (CH<sub>4</sub>) and 30-45% CO<sub>2</sub>.

**Q29. Which of the following fuels is utilized in thermal power generation to produce electricity?**

- (a) Coal
- (b) Petroleum fuel oils
- (c) Natural gas
- (d) All the above

**Correct Answer: Option (d)**

**Explanation:** Thermal power stations across the world and India burn coal (predominant in India), fuel oils, and natural gas to fire high-pressure steam boilers and gas turbines.

**Q30. Accelerated topsoil erosion across agricultural landscapes is driven by:**

- (a) Deforestation and clearing of vegetative cover
- (b) Faulty farming techniques and surface mining
- (c) Uncontrolled overgrazing by livestock
- (d) All the above

**Correct Answer: Option (d)**

**Explanation:** Vegetative root systems bind soil together. Removing tree canopy, overgrazing slopes, and leaving bare tilled land exposed to torrential rain causes catastrophic sheet and gully erosion.

**Q31. In which year did the United Nations host the historic Conference on the Human Environment at Stockholm?**

- (a) 1962
- (b) 1972
- (c) 1982
- (d) 1992

**Correct Answer: Option (b)**

**Explanation:** The Stockholm Conference was convened in June 1972, resulting in the Stockholm Declaration of 26 principles and the founding of UNEP.

**Q32. Which forest vegetation type naturally thrives in arid regions with low, erratic rainfall (< 70 cm/year)?**

- (a) Tropical Evergreen Forest
- (b) Temperate Coniferous Forest
- (c) Moist Deciduous Forest
- (d) Dry Tropical Thorn Forest

**Correct Answer: Option (d)**

**Explanation:** Dry tropical thorn forests (characteristic of Rajasthan, Gujarat, and parts of the Deccan plateau) feature xerophytic thorny trees (Acacia, Prosopis) adapted to semi-arid drought.

**Q33. Which fast-growing commercial exotic tree is noted for lowering local groundwater tables via massive transpiration?**

- (a) Coconut palm
- (b) Mango tree
- (c) Eucalyptus tree
- (d) Neem tree

**Correct Answer: Option (c)**

**Explanation:** Eucalyptus species possess extensive, deep root systems and high transpiration rates, often depleting surrounding aquifers when planted in semi-arid soils.

**Q34. In Indian drought categorization, agricultural famine conditions are precipitated when regional seasonal rainfall falls:**

- (a) Below 30% of normal
- (b) Below 40% of normal
- (c) Below 50% of normal
- (d) Below 60% of normal

**Correct Answer: Option (c)**

**Explanation:** A severe drought occurrence (rainfall deficiency > 50% below normal) triggers acute crop failure, dried aquifers, and agrarian distress leading to famine conditions.

**Q35. According to the India Meteorological Department (IMD), moderate drought is defined when rainfall deficiency is:**

- (a) 10-20% deficit
- (b) 26% to 50% deficit
- (c) 15-25% deficit
- (d) 5-10% deficit

**Correct Answer: Option (b)**

**Explanation:** The IMD classifies drought into: 'Moderate Drought' when seasonal rainfall deficit is between 26% and 50%, and 'Severe Drought' when deficit exceeds 50%.

**Q36. The steady rise in atmospheric carbon dioxide (CO<sub>2</sub>) concentrations is primarily attributable to:**

- (a) Deforestation and loss of planetary carbon sinks
- (b) Industrial fossil fuel combustion (coal, oil, gas)
- (c) Intensive vehicular emissions and energy production
- (d) All the above

**Correct Answer: Option (d)**

**Explanation:** Fossil fuel burning emits over 35 billion tonnes of CO<sub>2</sub> annually, while deforestation removes the photosynthetic terrestrial sinks that sequester carbon.

**Q37. Water is scientifically classified as a renewable natural resource because:**

- (a) It can be synthesized chemically in factories
- (b) It is consumed by living organisms without changing mass
- (c) It is constantly renewed, purified, and recharged by the solar-driven Hydrological Cycle
- (d) It has an unlimited infinite volume

**Correct Answer: Option (c)**

**Explanation:** The global hydrological cycle (evaporation, condensation, precipitation, surface runoff, infiltration) continuously recycles and redistributes freshwater across the globe.

**Q38. Acute freshwater scarcity in developing nations is driven predominantly by:**

- (a) Exponential population growth and urbanization
- (b) Over-exploitation and depletion of aquifers
- (c) Industrial pollution of available sources and unequal access
- (d) All the above

**Correct Answer: Option (d)**

**Explanation:** Water stress is an intersectional crisis: booming municipal/irrigation demand, relentless borewell over-extraction, and chemical poisoning of surface rivers.

**Q39. Which two forms of alcohol are commonly synthesized and utilized as liquid biofuels?**

- (a) Methanol and Ethanol
- (b) Propanol and Ethanol
- (c) Butanol and Glycerol
- (d) Hexanol and Methanol

**Correct Answer: Option (a)**

**Explanation:** Bioethanol (produced from yeast fermentation of sugarcane juice or corn starches) and Biomethanol are the two foremost alcohol-based renewable automotive fuel substitutes.

**Q40. Globally and within the Indian subcontinent, which natural disaster occurs most frequently and causes widespread loss?**

- (a) Coastal Tsunami
- (b) Volcanic Eruption
- (c) Deep Earthquake
- (d) Riverine Flood

**Correct Answer: Option (d)**

**Explanation:** Floods are the most common and damaging natural disaster, accounting for over 40% of all disaster events and affecting millions along the Ganga, Brahmaputra, and coastal river plains.

**Q41. How many terrestrial and marine Biodiversity Hotspots are officially recognized across the world today?**

- (a) 36 Hotspots
- (b) 25 Hotspots
- (c) 18 Hotspots
- (d) 50 Hotspots

**Correct Answer: Option (a)**

**Explanation:** Originally conceptualized by Norman Myers with 25 regions, Conservation International currently recognizes 36 global biodiversity hotspots that harbor 50% of the world's endemic plant species.

**Q42. United Nations World Water Day is celebrated each year on:**

- (a) March 12
- (b) March 22
- (c) April 22
- (d) June 5

**Correct Answer: Option (b)**

**Explanation:** World Water Day has been observed on March 22nd since 1993, highlighting the critical importance of freshwater conservation and advocating for the sustainable management of freshwater resources.

**Q43. Which Indian state became the first to make rooftop Rainwater Harvesting mandatory for all residential/commercial buildings?**

- (a) Tamil Nadu
- (b) Maharashtra
- (c) Kerala
- (d) Rajasthan

**Correct Answer: Option (a)**

**Explanation:** In 2001, the Government of Tamil Nadu under Chief Minister J. Jayalalithaa enacted landmark legislation making rooftop rainwater harvesting structures compulsory for every building.

**Q44. The primary drivers of widespread deforestation across tropical developing nations include:**

- (a) Shifting (Jhum) cultivation and agricultural expansion
- (b) Massive demand for industrial commercial timber and fuelwood
- (c) Overgrazing by pastoral livestock and unmanaged forest fires
- (d) All the above

**Correct Answer: Option (d)**

**Explanation:** Tropical deforestation is driven by agricultural frontier expansion, infrastructural road and dam building, commercial timber exploitation, and frequent anthropogenic wildfires.

**Q45. The high-temperature thermal oxidation process used to destroy non-biodegradable and biomedical waste is called:**

- (a) Aerobic Composting
- (b) Incineration
- (c) Mechanical Segregation
- (d) Sanitary Landfilling

**Correct Answer: Option (b)**

**Explanation:** Incineration destroys combustible solids at controlled temperatures (850°C-1200°C), converting waste into flue gases and inert ash while completely sterilizing biohazardous pathogens.

**Q46. The terminal inactivation and destruction of pathogenic microorganisms in municipal wastewater is accomplished via:**

- (a) Pre-screening bars
- (b) Extended aeration
- (c) Chemical Disinfection (Chlorination / UV)
- (d) Primary grit settling

**Correct Answer: Option (c)**

**Explanation:** Disinfection is the final barrier in wastewater treatment. Chlorination, ozonation, or ultraviolet (UV) irradiation disrupts microbial DNA and cell walls, destroying enteric pathogens.

**Q47. Discarded electrical, electronic, and IT equipment nearing the end of its useful life is classified as:**

- (a) Solid Waste
- (b) Domestic Trash
- (c) Organic Residue
- (d) e-waste

**Correct Answer: Option (d)**

**Explanation:** e-waste (Waste Electrical and Electronic Equipment, WEEE) contains both precious recoverable metals (gold, copper) and dangerous neurotoxic elements (lead, mercury, brominated flame retardants).

**Q48. The primary hydrological and ecological objective of Rooftop Rainwater Harvesting is to:**

- (a) Irrigate public lawns
- (b) Fill recreational surface ponds
- (c) Flush municipal drainage lines
- (d) Artificial recharge of the subterranean Groundwater Table

**Correct Answer: Option (d)**

**Explanation:** Channeling rooftop rain runoff through filtration media directly into dry open wells or recharge borewells artificially replenishes deep depleted aquifers, reversing local water table decline.

**Q49. Which of the following rocks/minerals represent essential non-metallic structural resources?**

- (a) Coal, silica, clay, cement
- (b) Iron, copper, aluminium, zinc
- (c) Gold, platinum, silver, titanium
- (d) Granite, limestone, marble

**Correct Answer: Option (d)**

**Explanation:** Granite, limestone, and marble are classic non-metallic industrial rocks used in construction. Limestone is also the foundational raw material for Portland cement manufacturing.

**Q50. Which layer of Earth's atmosphere contains the natural ozone shield that absorbs lethal UV radiation?**

- (a) Troposphere
- (b) Mesosphere
- (c) Stratosphere
- (d) Thermosphere

**Correct Answer: Option (c)**

**Explanation:** The Stratosphere (between 15 and 50 km above sea level) contains over 90% of atmospheric ozone. Solar UV-C and UV-B are absorbed here, protecting life on the planet.

## SECTION 6: FULLY SOLVED 2022 EXAMINATION PAPER (SET C)

**Paper:** Bachelor of Engineering Examination, 2022 (1st Year, 2nd Semester, Compulsory EVS, Code: C)

**Full Marks:** 100 | **Questions:** 50 MCQs (2 Marks each) | **Negative Marking:** NIL

**Q1. Which one of the following is NOT a standard chemical/physical characteristic measured for municipal solid waste?**

- (a) pH
- (b) N, P, K content
- (c) Carbohydrate content
- (d) Carbon-to-Nitrogen (C/N) ratio

**Correct Answer: Option (c)**

**Explanation:** Solid waste characterization measures proximate analysis (moisture, volatile matter, fixed carbon, ash), ultimate analysis (C, H, O, N, S), calorific value, NPK, and C/N ratio. Total carbohydrate is not a standard lumped design parameter.

**Q2. The best thermal disposal method for solid waste to guarantee complete destruction of pathogenic microorganisms is:**

- (a) Incineration
- (b) Sanitary Landfill
- (c) Windrow Composting
- (d) Pulverization

**Correct Answer: Option (a)**

**Explanation:** Incineration at high temperatures ( $> 850^{\circ}\text{C}$  to  $1100^{\circ}\text{C}$ ) completely oxidizes organic matter and ensures 100% thermal sterilization, making it mandatory for biomedical and clinical wastes.

**Q3. Which standard secondary biological method is most widely applied for large-scale municipal wastewater treatment?**

- (a) Biological aeration
- (b) Activated Sludge Process (ASP)
- (c) Anaerobic digestion
- (d) None of them

**Correct Answer: Option (b)**

**Explanation:** The Activated Sludge Process is the worldwide standard secondary wastewater treatment system. It circulates an active biomass of aerobic microorganisms to bio-oxidize soluble organic matter.

**Q4. When alum coagulant is added to wastewater containing natural calcium bicarbonate alkalinity, which compound forms as a heavy precipitate?**

- (a)  $\text{Al}_2(\text{SO}_4)_3$
- (b)  $\text{CaCO}_3$
- (c)  $\text{Al}(\text{OH})_3$
- (d)  $\text{Ca}(\text{OH})_2$

**Correct Answer: Option (c)**

**Explanation:** Alum  $[\text{Al}_2(\text{SO}_4)_3 \cdot 18\text{H}_2\text{O}]$  reacts with calcium bicarbonate alkalinity:  $\text{Al}_2(\text{SO}_4)_3 + 3\text{Ca}(\text{HCO}_3)_2 \rightarrow 2\text{Al}(\text{OH})_3$  (precipitate) +  $3\text{CaSO}_4 + 6\text{CO}_2$ . The gelatinous Aluminium Hydroxide floc sweeps suspended particles.

**Q5. How many distinct chemical forms of nitrogen typically exist in raw and treated domestic wastewater?**

- (a) 5 forms
- (b) 4 forms
- (c) 3 forms
- (d) 2 forms

**Correct Answer: Option (b)**

**Explanation:** Wastewater contains 4 primary forms of nitrogen: 1) Organic Nitrogen (protein/urea), 2) Ammoniacal Nitrogen ( $\text{NH}_3/\text{NH}_4^+$ ), 3) Nitrite Nitrogen ( $\text{NO}_2^-$ ), and 4) Nitrate Nitrogen ( $\text{NO}_3^-$ ).

**Q6. The controlled planned application of treated wastewater onto soil and land surfaces to achieve further purification is termed:**

- (a) Soil treatment
- (b) Land treatment
- (c) Wastewater treatment
- (d) Water treatment

**Correct Answer: Option (b)**

**Explanation:** Land treatment (including slow-rate irrigation, overland flow, and rapid infiltration) utilizes natural physical, chemical, and soil microbiological mechanisms to polish wastewater.

**Q7. Profuse proliferation of aquatic micro-vegetation that discolors surface water bodies and produces scum is termed:**

- (a) Algal bloom
- (b) Eutrophication
- (c) Fermentation
- (d) Both (a) and (b)

**Correct Answer: Option (d)**

**Explanation:** Eutrophication is the overarching process of nutrient over-enrichment, and the resulting explosive visible population surge of phytoplankton is called an 'Algal Bloom'.

**Q8. Natural processes such as biological anaerobic decay, forest wildfires, and volcanic eruptions are natural sources of:**

- (a) Gaseous pollutants
- (b) Water pollutants
- (c) Human-caused pollution
- (d) All of the above

**Correct Answer: Option (a)**

**Explanation:** Volcanoes emit SO<sub>2</sub> and CO<sub>2</sub>; forest fires release CO and particulates; wetlands and biological decay release CH<sub>4</sub> and H<sub>2</sub>S. These represent primary natural gaseous emissions.

**Q9. Which mechanical process is routinely employed at the head of a sewage treatment plant to lift wastewater and gain sufficient hydraulic head?**

- (a) Fermentation
- (b) Oxidation
- (c) Screening
- (d) Pumping

**Correct Answer: Option (d)**

**Explanation:** Raw sewage enters treatment plants at low gravity elevations. Wastewater lift pumping stations pump sewage to elevated headworks so it can flow by gravity through subsequent treatment units.

**Q10. The international Ramsar Convention (signed in Ramsar, Iran in 1971) is dedicated to the conservation of:**

- (a) Temperate Forests
- (b) Wetlands
- (c) Open Oceans
- (d) Wildlife

**Correct Answer: Option (b)**

**Explanation:** The Ramsar Convention is an intergovernmental treaty that provides the framework for national action and international cooperation for the conservation and wise use of wetlands.

**Q11. The biological conservation of plant and animal genetic resources directly within their natural native habitats is termed:**

- (a) Zoological garden
- (b) Ex-situ conservation
- (c) In-situ conservation
- (d) In-vivo conservation

**Correct Answer: Option (c)**

**Explanation:** In-situ conservation protects species within their natural ecosystems (National Parks, Sanctuaries, Biosphere Reserves, Sacred Groves). Ex-situ protects them outside (zoos, seed banks).

**Q12. Which extremely hazardous gas leaked into the atmosphere during the tragic 1984 Bhopal Disaster?**

- (a) Methyl isocyanate (MIC)
- (b) Potassium isothiocyanate
- (c) Ethyl isocyanate
- (d) Sodium isothiocyanate

**Correct Answer: Option (a)**

**Explanation:** Methyl isocyanate (MIC,  $\text{CH}_3\text{-N}=\text{C}=\text{O}$ ) leaked from the Union Carbide India Limited pesticide plant on December 2-3, 1984, reacting with water in lung tissues to produce devastating pulmonary edema.

**Q13. Which of the following is classified as a non-point (diffuse) source of surface water pollution?**

- (a) Industrial factory discharge pipes
- (b) Municipal sewage outfalls
- (c) Runoff from urban and suburban agricultural lands
- (d) All of these

**Correct Answer: Option (c)**

**Explanation:** Point sources enter waterways via a discrete conveyance pipe (factories, sewage treatment plants). Non-point sources arise from diffuse landscape runoff (farms, streets, construction sites).

**Q14. Which of the following cultivated fruit species possesses the highest genetic diversity in India?**

- (a) Teak
- (b) Tea
- (c) Mango
- (d) Potato

**Correct Answer: Option (c)**

**Explanation:** India is the primary centre of origin for Mango (*Mangifera indica*), boasting over 1,000 distinct cultivars and landraces showing enormous genetic variations.

**Q15. When the very last surviving individual of a biological species dies worldwide, that species has reached:**

- (a) Speciation
- (b) Extinction
- (c) Phylogenetic diversity
- (d) Adaptation

**Correct Answer: Option (b)**

**Explanation:** Extinction is the terminal, irreversible demise of a biological taxon when no living individuals remain anywhere on Earth (e.g., Passenger Pigeon, Dodo, Tasmanian Tiger).

**Q16. Which of the following soil classes exhibits the smallest pore size and lowest permeability?**

- (a) Sandy loam
- (b) Peat soil
- (c) Clayey soil
- (d) Gravel

**Correct Answer: Option (c)**

**Explanation:** Clay particles are colloidal ( $< 0.002$  mm). Although total porosity is high, the microscopic pore diameter creates high surface tension and hydraulic friction, making clay the least permeable soil.

**Q17. Agricultural lime (Calcium carbonate / Calcium hydroxide) is routinely added to neutralize \_\_\_\_\_ soils:**

- (a) Acidic
- (b) Dry
- (c) Salty
- (d) Alkaline

**Correct Answer: Option (a)**

**Explanation:** Agricultural liming raises soil pH in acidic soils ( $\text{pH} < 6.0$ ), neutralizing toxic aluminium/manganese ions and restoring nutrient availability (phosphorus, calcium, magnesium).

**Q18. Which globally recognized Biodiversity Hotspot is located predominantly in Southern and Western India?**

- (a) Indo-Gangetic plain
- (b) Sundarbans delta
- (c) Eastern Ghats
- (d) Western Ghats

**Correct Answer: Option (d)**

**Explanation:** The Western Ghats mountain chain along India's western coast is one of the world's 8 'hottest hotspots' of biodiversity, harboring extraordinary endemic flora, amphibians, and mammals.

**Q19. Sacred Groves (deorais / orans / sarans) maintained by indigenous tribal communities are invaluable for:**

- (a) Commercial timber extraction
- (b) Conserving endangered, rare, and relict species in virgin habitat
- (c) Industrial grazing
- (d) Maximizing soil erosion

**Correct Answer: Option (b)**

**Explanation:** Sacred groves are patches of pristine forest protected through traditional religious beliefs, acting as vital refugia for rare medicinal herbs, relict plants, and endemic fauna.

**Q20. The historic Chipko Movement in the Garhwal Himalayas was famously strengthened and expanded under the leadership of:**

- (a) Amrita Devi Bishnoi
- (b) Medha Patkar
- (c) Dr. A. K. Banerjee
- (d) Sunder Lal Bahuguna

**Correct Answer: Option (d)**

**Explanation:** Gandhian environmentalist Sunder Lal Bahuguna popularized the Chipko slogan 'Ecology is permanent economy', undertaking long foot marches to ban commercial felling in the Himalayas.

**Q21. Balancing the present societal need to utilize natural resources while conserving them for future generations is defined as:**

- (a) Sustainable development
- (b) Resource exhaustion
- (c) Commercial exploitation
- (d) Environmental neglect

**Correct Answer: Option (a)**

**Explanation:** The 1987 Brundtland Commission defined Sustainable Development: 'Development that meets the needs of the present without compromising the ability of future generations to meet their own needs'.

**Q22. The internationally recognized Red Data Book compiles exhaustive conservation information regarding:**

- (a) All agricultural crop plants
- (b) All domestic animal breeds
- (c) Globally threatened and endangered biological species
- (d) Economically valuable timber species

**Correct Answer: Option (c)**

**Explanation:** Maintained and published by the IUCN, the Red Data Book evaluates species extinction vulnerability across standardized risk categories (Vulnerable, Endangered, Critically Endangered).

**Q23. The Central Pollution Control Board (CPCB) was first constituted by the Government of India in the year:**

- (a) 1947
- (b) 1950
- (c) 1974
- (d) 1980

**Correct Answer: Option (c)**

**Explanation:** The CPCB was established as a statutory organization in September 1974 under Section 3 of the Water (Prevention and Control of Pollution) Act, 1974.

**Q24. Which two major nations together account for approximately 57% of all large operational dams globally?**

- (a) India and China
- (b) India and Brazil
- (c) China and Brazil
- (d) India and Japan

**Correct Answer: Option (a)**

**Explanation:** According to the World Commission on Dams (WCD), China has over 23,800 large dams and India has over 5,100 large dams; together they hold roughly 57% of the world's mega-dams.

**Q25. Which of the following mineral groups represents essential non-metallic industrial and building resources?**

- (a) Granite, marble, limestone
- (b) Gold, silver, platinum
- (c) Iron, aluminum, zinc, copper
- (d) Lead, mercury, nickel, tin

**Correct Answer: Option (a)**

**Explanation:** Granite, marble, and limestone are non-metallic structural minerals. Gold and iron are metallic; coal and petroleum are classified specifically as mineral fuels.

**Q26. The historic 1992 'Earth Summit' convened in Rio de Janeiro is formally associated with:**

- (a) Adoption of the Convention on Biological Diversity (CBD) and Agenda 21
- (b) Kyoto Protocol 1997
- (c) Montreal Protocol 1987
- (d) Geneva Convention

**Correct Answer: Option (a)**

**Explanation:** The United Nations Conference on Environment and Development (UNCED / Rio Earth Summit 1992) produced the Rio Declaration, Agenda 21, the CBD, and the UNFCCC.

**Q27. In India, Environmental Impact Assessment (EIA) was made statutory and mandatory under which regulation?**

- (a) Indian Forest Act 1927
- (b) Environment (Protection) Act, 1986
- (c) Water Act 1974
- (d) Wildlife Act 1972

**Correct Answer: Option (b)**

**Explanation:** The Ministry of Environment, Forest and Climate Change (MoEFCC) notified the first statutory EIA Notification in January 1994 under Section 3 of the Environment (Protection) Act, 1986.

**Q28. CNG (Compressed Natural Gas) is environmentally characterized as a:**

- (a) Highly polluting fuel
- (b) Clean-burning automotive fuel
- (c) Harmful toxic fuel
- (d) Radioactive fuel

**Correct Answer: Option (b)**

**Explanation:** Composed primarily of methane (CH<sub>4</sub>), CNG burns with virtually zero particulate matter, lower carbon monoxide, and significantly lower NO<sub>x</sub> compared to diesel and petrol.

**Q29. World Environment Day is observed across the world each year on:**

- (a) 1st June
- (b) 5th June
- (c) 7th June
- (d) 21st June

**Correct Answer: Option (b)**

**Explanation:** Celebrated annually on June 5th since 1973, it serves as the United Nations' primary vehicle for encouraging worldwide awareness and action for the environment.

**Q30. The comprehensive Wildlife Protection Act was officially enforced across India in the year:**

- (a) 1972
- (b) 1974
- (c) 1977
- (d) 1981

**Correct Answer: Option (a)**

**Explanation:** Enacted in 1972, this landmark legislation provided the legal backbone for wildlife conservation, anti-poaching penalties, and the creation of Protected Areas in India.

**Q31. The physical, geographical, and ecological place where a biological organism or community naturally lives is called its:**

- (a) Niche
- (b) Biome
- (c) Habitat
- (d) Habit

**Correct Answer: Option (c)**

**Explanation:** Habitat is the physical locality where an organism lives (its ecological 'address'). Ecological Niche, by contrast, refers to its functional role or 'profession' within that community.

**Q32. Which of the following represents a renewable, yet exhaustible natural energy resource?**

- (a) Coal
- (b) Petroleum
- (c) Kerosene
- (d) Biomass / Forest

**Correct Answer: Option (d)**

**Explanation:** Biomass renews itself naturally through plant growth. However, if human consumption and deforestation outpace biological growth rates, the resource is rapidly exhausted.

**Q33. Consider the food chain: Dead Animal carcass -> Blowfly maggots -> Frog -> Snake. This is a classic example of a:**

- (a) Detrital food chain
- (b) Grazing food chain
- (c) Parasitic food chain
- (d) Autotrophic food chain

**Correct Answer: Option (a)**

**Explanation:** A detritus food chain begins with dead organic matter (detritus), which is consumed by detritivores (decomposers/maggots), which are then preyed upon by secondary carnivores.

**Q34. Which of the following primary anthropogenic emission sources are responsible for the formation of SMOG?**

- (a) Emissions from industrial incinerators
- (b) Hydrocarbon and NO<sub>x</sub> emissions from motor vehicles
- (c) Both incinerator emissions and vehicular exhaust
- (d) None of the above

**Correct Answer: Option (c)**

**Explanation:** Smog (both industrial sulphurous smog and photochemical smog) results from combinations of soot, vehicular NO<sub>x</sub>, volatile hydrocarbons, and industrial incinerator flue gases.

**Q35. Which of the following atmospheric pollutants is classified specifically as a solid/liquid particulate pollutant?**

- (a) Ground-level Ozone
- (b) Radon gas
- (c) Fly Ash
- (d) Ethylene gas

**Correct Answer: Option (c)**

**Explanation:** Fly ash consists of microscopic unburned inorganic mineral particles carried out of coal-fired thermal power plant boilers with flue gases (captured by electrostatic precipitators).

**Q36. Which atmospheric air pollutant is primarily responsible for chemical corrosion and yellowing of the marble Taj Mahal?**

- (a) Elemental Sulphur
- (b) Chlorine gas
- (c) Sulphur dioxide (SO<sub>2</sub>)
- (d) Carbon monoxide

**Correct Answer: Option (c)**

**Explanation:** SO<sub>2</sub> emitted from the nearby Mathura Oil Refinery and local industries reacts with rain to form sulfuric acid, converting marble (CaCO<sub>3</sub>) into gypsum [CaSO<sub>4</sub>·2H<sub>2</sub>O], termed 'Marble Cancer'.

**Q37. Pesticide DDT residues and discarded Aluminium beverage cans are classic examples of:**

- (a) Rapidly degradable pollutants
- (b) Secondary pollutants
- (c) Biodegradable pollutants
- (d) Non-biodegradable persistent pollutants

**Correct Answer: Option (d)**

**Explanation:** Non-biodegradable pollutants cannot be metabolized or broken down by microbial action within ordinary timeframes; DDT persists for decades, and aluminum takes centuries to corrode.

**Q38. Which of the following atmospheric pollutant complexes is classified as secondary air pollutants?**

- (a) Photochemical Smog and Tropospheric Ozone
- (b) Sulphur trioxide (SO<sub>3</sub>)
- (c) Peroxyacetyl Nitrate (PAN)
- (d) All of the above

**Correct Answer: Option (d)**

**Explanation:** Secondary pollutants are not emitted directly; they are synthesized in the atmosphere through photochemical or thermal reactions among primary emissions (NO<sub>x</sub>, VOCs, SO<sub>2</sub>).

**Q39. Prolonged exposure to elevated urban ambient air pollution primarily precipitates:**

- (a) Soil erosion
- (b) Acidification of oceans
- (c) Severe chronic respiratory and cardiovascular illnesses
- (d) Landslides

**Correct Answer: Option (c)**

**Explanation:** Respirable suspended particulate matter (PM<sub>2.5</sub> and PM<sub>10</sub>) penetrates deep into alveolar pulmonary capillaries, triggering asthma, chronic obstructive pulmonary disease (COPD), and lung cancer.

**Q40. Which of the following atmospheric gases are potent greenhouse gases responsible for global warming?**

- (a) Methane (CH<sub>4</sub>)
- (b) Molecular Nitrogen (N<sub>2</sub>)
- (c) Argon
- (d) Both (a) and trace halocarbons

**Correct Answer: Option (a)**

**Explanation:** Methane (CH<sub>4</sub>) has a Global Warming Potential (GWP) 28 times greater than CO<sub>2</sub> over a 100-year horizon. Nitrogen (N<sub>2</sub>) and Argon have symmetrical diatomic/monatomic structures and do not absorb IR.

**Q41. Which of the following statements is scientifically accurate regarding CNG (Compressed Natural Gas)?**

- (a) It is an environmentally clean automotive alternative fuel
- (b) It is more toxic than diesel exhaust
- (c) It produces heavy particulate soot
- (d) It cannot be compressed safely

**Correct Answer: Option (a)**

**Explanation:** CNG is primarily methane. Being a light hydrocarbon with high H/C ratio, it undergoes clean combustion, producing minimal particulate soot, low CO, and reduced greenhouse gas intensity.

**Q42. Which dissolved inorganic chemical salt is a principal cause of PERMANENT hardness in water?**

- (a) Magnesium sulphate ( $\text{MgSO}_4$ )
- (b) Magnesium bicarbonate [ $\text{Mg}(\text{HCO}_3)_2$ ]
- (c) Calcium bicarbonate [ $\text{Ca}(\text{HCO}_3)_2$ ]
- (d) Sodium chloride

**Correct Answer: Option (a)**

**Explanation:** Temporary hardness is caused by bicarbonates of Ca and Mg (removable by boiling). Permanent hardness is caused by sulphates and chlorides of calcium and magnesium ( $\text{MgSO}_4$ ,  $\text{CaCl}_2$ ).

**Q43. Which specialized water treatment technique is employed to purge and measure volatile taste- and odour-causing organic compounds?**

- (a) Gravity settling
- (b) Back-flushing
- (c) Air Stripping
- (d) Coagulation

**Correct Answer: Option (c)**

**Explanation:** Air stripping passes large volumes of air through water, transferring volatile organic compounds (VOCs), hydrogen sulphide, and odour components from the liquid phase to the gas phase.

**Q44. In most freshwater aquatic lakes, the biological productivity of algae and macrophytes is limited primarily by the availability of:**

- (a) Inorganic Carbon
- (b) Atmospheric Nitrogen
- (c) Phosphorus (as dissolved orthophosphates)
- (d) Potassium

**Correct Answer: Option (c)**

**Explanation:** According to Liebig's Law of the Minimum, Phosphorus is the limiting nutrient in freshwater ecosystems ( $\text{N/P ratio} > 16$ ). Excess phosphorus influx triggers uncontrolled eutrophication.

**Q45. Which of the following human infectious diseases is NOT transmitted via the drinking of contaminated water?**

- (a) Measles
- (b) Typhoid fever
- (c) Cholera
- (d) Hepatitis A

**Correct Answer: Option (a)**

**Explanation:** Measles is an airborne viral disease spread via respiratory droplets (coughing/sneezing). Typhoid (*Salmonella typhi*), Cholera (*Vibrio cholerae*), and Hepatitis A are classic waterborne diseases.

**Q46. In environmental acoustics, 'Noise' is officially defined as:**

- (a) Any loud desirable musical sound
- (b) Desirable and wanted rhythmic acoustic vibration
- (c) Undesirable, unwanted, or disruptive sound that causes physiological annoyance
- (d) High-frequency ultrasound

**Correct Answer: Option (c)**

**Explanation:** Noise is defined as unwanted, unpleasant, or damaging sound that interferes with human communication, causes psychological stress, or impairs auditory health.

**Q47. What is the internationally standardized reference sound pressure level corresponding to the threshold of human hearing?**

- (a) 0 dB
- (b) 20 dB
- (c) 65 dB
- (d) 100 dB

**Correct Answer: Option (a)**

**Explanation:** 0 dB on the decibel scale represents the threshold of human hearing at 1000 Hz, corresponding to an acoustic sound pressure of 20 micropascals (20  $\mu\text{Pa}$ ).

**Q48. A temporary reversible hearing impairment resulting from short-term exposure to loud noise (> 90 dB) is termed:**

- (a) Temporary ear infection
- (b) Tinnitus vertigo
- (c) Temporary Threshold Shift (TTS)
- (d) Acoustic barotrauma

**Correct Answer: Option (c)**

**Explanation:** *Temporary Threshold Shift (auditory fatigue) is a reversible elevation of the hearing threshold caused by metabolic fatigue of outer hair cells in the cochlea after loud noise exposure.*

**Q49. Which of the following global ecological phenomena is NOT a direct adverse consequence of global warming?**

- (a) Accelerated mean sea-level rise
- (b) Direct increase in lethal Solar UV-B radiation reaching Earth surface
- (c) Widespread retreat of alpine glaciers and polar ice
- (d) Extreme, erratic weather patterns and intensified cyclonic activity

**Correct Answer: Option (b)**

**Explanation:** *Increased UV-B radiation is caused by Stratospheric Ozone Layer Depletion (via CFCs), NOT by global warming. Global warming is caused by greenhouse gas trapping of outgoing infrared heat.*

**Q50. Which category of solid hospital waste presents the greatest biological hazard of transmitting Hepatitis B, Hepatitis C, and HIV via percutaneous skin puncture?**

- (a) Discarded electronic IT waste
- (b) Biomedical Sharps waste (needles, scalpels, lancets)
- (c) Agricultural pesticide containers
- (d) Municipal food scrap waste

**Correct Answer: Option (b)**

**Explanation:** *Contaminated biomedical sharps (syringes, needles, surgical blades) pose direct percutaneous inoculation risks for blood-borne viral pathogens (HBV, HCV, HIV).*

## SECTION 7: CURATED HIGH-YIELD QUESTION BANK (LORETO COLLEGE DRILL)

**Source:** Loreto College Department of Geography (Compiled by Debasree Sinha) for CBCS AECC-2.  
These 40 high-probability questions are repeatedly drawn upon by university exam question setting boards.

**Q1. The landmark 1987 report entitled 'Our Common Future' by the Brundtland Commission pertains to:**

- (a) Biodiversity conservation
- (b) Sustainable Development
- (c) Nuclear disarmament
- (d) Space exploration

**Correct Answer: Option (b)**

**Explanation:** The UN World Commission on Environment and Development (chaired by Gro Harlem Brundtland) published 'Our Common Future', defining Sustainable Development.

**Q2. Joint Forest Management (JFM) was pioneered for the first time in India in 1971 at:**

- (a) Chhota Nagpur Plateau
- (b) Dandakaranya Forest
- (c) Arabari Forest Range (West Bengal)
- (d) Nilgiri Biosphere

**Correct Answer: Option (c)**

**Explanation:** JFM was pioneered in the Arabari forest range of Midnapore, West Bengal, by DFO Dr. A.K. Banerjee, revitalizing degraded Sal forests with community partnership.

**Q3. Freshwater flowing continuously through rivers and streams is ecologically classified as:**

- (a) Lotic water
- (b) Lentic water
- (c) Estuarine water
- (d) Oceanic water

**Correct Answer: Option (a)**

**Explanation:** Lotic ecosystems encompass flowing, running freshwaters (rivers, brooks, rapids). Lentic ecosystems encompass still, standing waters (ponds, lakes, bogs).

**Q4. The directional flow of energy through the trophic levels of an ecosystem is always:**

- (a) Unidirectional
- (b) Multidirectional
- (c) Cyclic
- (d) Reversible

**Correct Answer: Option (a)**

**Explanation:** Energy flow follows thermodynamic laws and Lindeman's 10% efficiency principle: solar energy captured by autotrophs moves unidirectionally to consumers and dissipates as metabolic heat.

**Q5. The scientific term 'Ecosystem' was first coined and introduced by:**

- (a) E.W. Zimmermann
- (b) A.G. Tansley
- (c) Charles Elton
- (d) Eugene Odum

**Correct Answer: Option (b)**

**Explanation:** British ecologist Sir Arthur George Tansley coined the term 'ecosystem' in 1935 to describe the integrated system of biotic community and physical abiotic environment.

**Q6. Free-floating or weakly swimming microscopic photosynthetic organisms in water bodies are called:**

- (a) Nekton
- (b) Periphyton
- (c) Plankton (Phytoplankton)
- (d) Benthos

**Correct Answer: Option (c)**

**Explanation:** Plankton drift with water currents (phytoplankton are microscopic plants, zooplankton are microscopic animals). Nekton are active swimmers (fish); benthos live on the bottom.

**Q7. Sundarbans and Buxa in West Bengal are officially designated as:**

- (a) Tiger Reserves
- (b) Biosphere Reserves
- (c) Bird Sanctuaries
- (d) Ramsar Sites

**Correct Answer: Option (a)**

**Explanation:** Both Sundarbans (mangrove ecosystem in the Ganga-Brahmaputra delta) and Buxa (in Alipurduar district) are designated Tiger Reserves under Project Tiger.

**Q8. India's first commercial atomic nuclear power station was commissioned at:**

- (a) Rawatbhata (Rajasthan)
- (b) Kalpakkam (Tamil Nadu)
- (c) Tarapur (Maharashtra)
- (d) Narora (Uttar Pradesh)

**Correct Answer: Option (c)**

**Explanation:** Tarapur Atomic Power Station (TAPS) in Palghar district, Maharashtra, was India's first commercial nuclear power facility, commissioned in October 1969.

**Q9. The historic Chipko Movement in Uttarakhand is associated with opposition to commercial felling and:**

- (a) Sardar Sarovar Project
- (b) Silent Valley Project
- (c) Tehri Dam Project
- (d) Narmada Project

**Correct Answer: Option (c)**

**Explanation:** Chipko activists, notably Sunderlal Bahuguna, were also prominent leaders in protesting the construction and seismic vulnerability of the massive Tehri Dam.

**Q10. International Earth Day is observed annually across the world on:**

- (a) April 22
- (b) March 22
- (c) June 5
- (d) September 16

**Correct Answer: Option (a)**

**Explanation:** Earth Day was first celebrated on April 22, 1970, founded by US Senator Gaylord Nelson, and has grown into a global civic mobilization for environmental action.

**Q11. The famous environmental protest movement to save the pristine Silent Valley rainforest took place in:**

- (a) Karnataka
- (b) Kerala
- (c) Tamil Nadu
- (d) Andhra Pradesh

**Correct Answer: Option (b)**

**Explanation:** Silent Valley Movement was launched in 1978 in Palakkad, Kerala, led by the Kerala Sastra Sahitya Parishad (KSSP) to save the lion-tailed macaque from a hydro project.

**Q12. The physical magnitude of energy released by an earthquake is measured on the:**

- (a) Mercalli scale
- (b) Richter scale
- (c) Beaufort scale
- (d) Decibel scale

**Correct Answer: Option (b)**

**Explanation:** The Richter magnitude scale is a logarithmic scale quantifying the total seismic energy released at an earthquake's subterranean focal point.

**Q13. The United Nations Conference on Environment and Development (Earth Summit) was held in 1992 at:**

- (a) Stockholm, Sweden
- (b) Kyoto, Japan
- (c) Rio de Janeiro, Brazil
- (d) Paris, France

**Correct Answer: Option (c)**

**Explanation:** The Rio Earth Summit (June 1992) adopted Agenda 21, the Rio Declaration, the Convention on Biological Diversity (CBD), and the UNFCCC.

**Q14. Which of the following gases is NOT a greenhouse gas?**

- (a) Carbon dioxide (CO<sub>2</sub>)
- (b) Methane (CH<sub>4</sub>)
- (c) Nitrogen gas (N<sub>2</sub>)
- (d) Nitrous oxide (N<sub>2</sub>O)

**Correct Answer: Option (c)**

**Explanation:** Molecular nitrogen (N<sub>2</sub>) comprises 78% of dry air. Because of its symmetric non-polar homonuclear diatomic structure, it does not absorb infrared radiation and is NOT a greenhouse gas.

**Q15. In water quality engineering, BOD and COD parameters evaluate:**

- (a) Air pollution potential
- (b) Water pollution & organic load
- (c) Soil fertility index
- (d) Nuclear radioactivity

**Correct Answer: Option (b)**

**Explanation:** BOD (Biochemical Oxygen Demand) and COD (Chemical Oxygen Demand) quantify the amount of dissolved oxygen consumed to oxidize organic pollutants in water.

**Q16. The toxic industrial gas responsible for thousands of fatalities in the 1984 Bhopal Gas Tragedy was:**

- (a) Phosgene
- (b) Hydrogen cyanide
- (c) Methyl Isocyanate (MIC)
- (d) Carbon monoxide

**Correct Answer: Option (c)**

**Explanation:** Methyl Isocyanate (MIC) leaked from tank E610 of the Union Carbide plant in Bhopal on the night of December 2-3, 1984.

**Q17. Stagnant, motionless water bodies such as lakes, ponds, and swamps are termed:**

- (a) Lotic ecosystems
- (b) Lentic ecosystems
- (c) Estuarine ecosystems
- (d) Marine ecosystems

**Correct Answer: Option (b)**

**Explanation:** Lentic (from Latin 'lentus', meaning slow or motionless) refers to calm, standing inland freshwater systems.

**Q18. Severe bone softening, osteomalacia, and painful joint deformities in 'Itai-Itai disease' are caused by poisoning from:**

- (a) Lead (Pb)
- (b) Cadmium (Cd)
- (c) Mercury (Hg)
- (d) Arsenic (As)

**Correct Answer: Option (b)**

**Explanation:** Itai-Itai ('it hurts-it hurts') disease occurred in Toyama Prefecture, Japan, caused by cadmium pollution released into rivers from mining operations, contaminating downstream rice fields.

**Q19. Neurological sensory impairment and congenital brain damage in 'Minamata disease' was caused by industrial poisoning from:**

- (a) Methylmercury (Hg)
- (b) Cadmium (Cd)
- (c) Arsenic (As)
- (d) Fluoride (F)

**Correct Answer: Option (a)**

**Explanation:** Minamata disease was first identified in 1956 in Minamata Bay, Japan, caused by bioaccumulation of industrial methylmercury in fish eaten by local communities.

**Q20. Unpolluted natural rainwater has a pH of approximately 5.6. Rainwater is defined as 'Acid Rain' when its pH falls:**

- (a) Below 5.6
- (b) Below 7.0
- (c) Below 8.0
- (d) Exactly 7.0

**Correct Answer: Option (a)**

**Explanation:** Because natural dissolved CO<sub>2</sub> yields a weakly acidic pH of 5.6, any precipitation with pH < 5.6 (driven by strong acids H<sub>2</sub>SO<sub>4</sub> and HNO<sub>3</sub>) is classified as acid rain.

**Q21. How many internationally recognized Biodiversity Hotspots cover parts of India?**

- (a) 2 Hotspots
- (b) 4 Hotspots
- (c) 8 Hotspots
- (d) 10 Hotspots

**Correct Answer: Option (b)**

**Explanation:** Four global biodiversity hotspots extend into India: 1) The Western Ghats, 2) The Eastern Himalayas, 3) Indo-Burma (Northeastern states), and 4) Sundaland (Nicobar Islands).

**Q22. The international body IUCN officially stands for:**

- (a) International Union for Conservation of Nature and Natural Resources
- (b) Indian Union for Conservation of Nature
- (c) International United Conservation Network
- (d) Intergovernmental Union for Climate and Nature

**Correct Answer: Option (a)**

**Explanation:** IUCN was founded in October 1948 in Fontainebleau, France. Its mission is to conserve nature and promote sustainable use of natural resources.

**Q23. Project Tiger, India's landmark national campaign to save the Bengal tiger from extinction, was launched in:**

- (a) 1970
- (b) 1973
- (c) 1980
- (d) 1985

**Correct Answer: Option (b)**

**Explanation:** Project Tiger was launched on April 1, 1973, from Corbett National Park under Prime Minister Indira Gandhi's tenure, creating an initial network of 9 Tiger Reserves.

**Q24. The Appiko Movement, a grassroots forest protection movement inspired by Chipko, took place in:**

- (a) Garhwal, Uttarakhand
- (b) Western Ghats of Karnataka
- (c) Aravalli hills, Rajasthan
- (d) Satpura hills, MP

**Correct Answer: Option (b)**

**Explanation:** In 1983, villagers led by Panduranga Hegde launched the Appiko movement ('Appiko' means to embrace in Kannada) in the Uttara Kannada district of Karnataka to protect indigenous forests.

**Q25. The landmark book 'Silent Spring' (1962), which catalyzed the modern global environmental movement, was authored by:**

- (a) Rachel Carson
- (b) Paul Ehrlich
- (c) Aldo Leopold
- (d) E.O. Wilson

**Correct Answer: Option (a)**

**Explanation:** Marine biologist Rachel Carson wrote 'Silent Spring', meticulously detailing how indiscriminate synthetic pesticide use (notably DDT) devastated avian populations and contaminated ecosystems.

**Q26. The international agreement aimed at freezing and phasing out production of Ozone Depleting Substances is the:**

- (a) Kyoto Protocol
- (b) Montreal Protocol
- (c) Paris Agreement
- (d) Basel Convention

**Correct Answer: Option (b)**

**Explanation:** Signed on September 16, 1987, in Montreal, Canada, the Montreal Protocol is widely recognized as the most universally successful multilateral environmental treaty in history.

**Q27. The comprehensive legal framework 'Environment (Protection) Act' was enacted by Indian Parliament in:**

- (a) 1972
- (b) 1974
- (c) 1986
- (d) 1991

**Correct Answer: Option (c)**

**Explanation:** Enacted under Article 253 of the Constitution in May 1986 in the immediate wake of the Bhopal Gas Disaster, it serves as India's foundational 'umbrella' environmental law.

**Q28. Disposal of civil environmental cases and enforcement of legal rights relating to environment in India is handled by the:**

- (a) Supreme Court only
- (b) Central Vigilance Commission
- (c) National Green Tribunal (NGT)
- (d) Planning Commission

**Correct Answer: Option (c)**

**Explanation:** Established in 2010 under the NGT Act, the National Green Tribunal is a specialized fast-track judicial body mandated to dispose of environmental disputes within 6 months.

**Q29. The catastrophic 1986 nuclear reactor core meltdown and explosion occurred at:**

- (a) Three Mile Island (USA)
- (b) Fukushima Daiichi (Japan)
- (c) Chernobyl (Ukraine/USSR)
- (d) Kashiwazaki (Japan)

**Correct Answer: Option (c)**

**Explanation:** On April 26, 1986, Reactor No. 4 at the Chernobyl Nuclear Power Plant in Ukraine suffered a catastrophic steam explosion and open graphite fire, releasing massive radioactive isotopes.

**Q30. In any terrestrial or aquatic ecological food chain, Trophic Level 1 (T1) is universally occupied by:**

- (a) Herbivores (Primary consumers)
- (b) Autotrophic Green Plants / Producers
- (c) Top Carnivores
- (d) Saprophytic Fungi

**Correct Answer: Option (b)**

**Explanation:** Trophic Level 1 is always formed by photoautotrophs (plants, algae, cyanobacteria) that convert solar photon energy into chemical bond energy through photosynthesis.

**Q31. Dr. Salim Ali (1896-1987), known as the 'Birdman of India', was an internationally celebrated:**

- (a) Marine Biologist
- (b) Ornithologist and Naturalist
- (c) Nuclear Physicist
- (d) Forest Commissioner

**Correct Answer: Option (b)**

**Explanation:** Salim Ali conducted pioneering systematic bird surveys across India, wrote authoritative field guides, and penned the famous autobiography 'The Fall of a Sparrow'.

**Q32. The large multipurpose Sardar Sarovar Dam is constructed across which major west-flowing Indian river?**

- (a) Tapi River
- (b) Narmada River
- (c) Godavari River
- (d) Mahanadi River

**Correct Answer: Option (b)**

**Explanation:** Built on the Narmada River in Gujarat, the Sardar Sarovar Dam triggered major resistance led by Medha Patkar and Baba Amte under the 'Narmada Bachao Andolan' over displacement issues.

**Q33. The Tehri Dam, the highest rock-and-earth fill dam in India (260 m), is constructed on the:**

- (a) Yamuna River
- (b) Alaknanda River
- (c) Bhagirathi River
- (d) Sutlej River

**Correct Answer: Option (c)**

**Explanation:** Located near Tehri in Garhwal, Uttarakhand, the dam harnesses the Bhagirathi River (a primary headstream of the sacred Ganga).

**Q34. Which of the following endemic Indian wildlife species is officially categorized as Critically Endangered?**

- (a) Great Indian Bustard
- (b) Common House Crow
- (c) Rhesus Macaque
- (d) Indian Peafowl

**Correct Answer: Option (a)**

**Explanation:** The Great Indian Bustard (*Ardeotis nigriceps*), native to arid grasslands of Rajasthan and Gujarat, has declined to fewer than 150 individuals due to hunting and collisions with high-voltage power lines.

**Q35. Which Indian state harbors the highest percentage of its geographical area under forest cover?**

- (a) Madhya Pradesh
- (b) Mizoram
- (c) Arunachal Pradesh
- (d) Odisha

**Correct Answer: Option (b)**

**Explanation:** According to the India State of Forest Report (ISFR), Mizoram has the highest forest cover as a percentage of total state area (> 84%). By absolute forest area, Madhya Pradesh ranks first.

**Q36. Which insect vector is biologically responsible for the transmission of the parasitic disease Leishmaniasis (Kala-azar)?**

- (a) Tsetse fly
- (b) Sandfly (Phlebotomus)
- (c) Aedes mosquito
- (d) Blackfly

**Correct Answer: Option (b)**

**Explanation:** Visceral leishmaniasis (Kala-azar) is caused by the protozoan parasite *Leishmania donovani* and transmitted by the bite of infected female *Phlebotomus* sandflies.

**Q37. Under the Constitution of India, which Directive Principle (Article) directs the State to protect and improve the environment?**

- (a) Article 21
- (b) Article 48A
- (c) Article 51A(g)
- (d) Article 14

**Correct Answer: Option (b)**

**Explanation:** Article 48A (added via 42nd Amendment, 1976) states: 'The State shall endeavour to protect and improve the environment and to safeguard the forests and wild life of the country'.

**Q38. Article 51A(g) of the Constitution of India establishes that it is the Fundamental Duty of every citizen of India to:**

- (a) Pay environmental taxes
- (b) Protect and improve the natural environment including forests, lakes, rivers and wildlife
- (c) Plant at least ten trees per year
- (d) Build rainwater harvesting structures

**Correct Answer: Option (b)**

**Explanation:** Article 51A(g) mandates a fundamental duty for every Indian citizen to cherish, protect, and improve the natural environment and have compassion for living creatures.

**Q39. The book 'Staying Alive: Women, Ecology and Survival in India' was authored by prominent ecofeminist:**

- (a) Arundhati Roy
- (b) Dr. Vandana Shiva
- (c) Medha Patkar
- (d) Mahasweta Devi

**Correct Answer: Option (b)**

**Explanation:** Dr. Vandana Shiva authored 'Staying Alive' (1988), articulating the philosophical foundations of ecofeminism, indigenous seed conservation (Navdanya), and peasant struggles.

**Q40. What is the designated apex statutory authority and policy-making body in India for environmental protection and climate governance?**

- (a) Central Pollution Control Board (CPCB)
- (b) Ministry of Environment, Forest and Climate Change (MoEFCC)
- (c) National Green Tribunal (NGT)
- (d) Wildlife Crime Control Bureau

**Correct Answer: Option (b)**

**Explanation:** The Ministry of Environment, Forest and Climate Change (MoEFCC) is the nodal executive ministry responsible for national environmental planning, laws, policies, and international negotiations.

## SECTION 8: FINAL HOUR EXAM HALL PROTOCOL & OMR STRATEGY

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**1. OMR Code Verification:** Immediately verify that your Question Booklet Code (A, B, C, or D) is correctly darkened in the OMR series bubble. An error here leads to electronic scanner misgrading.

**2. Strict 3-Pass Solving Strategy:**

- **Pass 1 (0 to 45 mins):** Read through all 50 questions rapidly. Solve the 30-35 questions that you know with 100% certainty (direct factual questions on dates, acts, national parks, chemical names). Shade them neatly.
- **Pass 2 (45 to 90 mins):** Tackle analytical or calculation questions (BOD/COD relations, wastewater treatment stages, atmospheric layers, direct vs indirect forest products). Use the process of elimination to eliminate 2 wrong options.
- **Pass 3 (90 to 110 mins): THE ZERO BLANK PROTOCOL.** Check for any unshaded circles. Because there is NO negative marking, leaving even one bubble blank is giving away free marks. For completely unknown questions, use informed elimination or pick Option B or C.

**3. High-Frequency Pitfall Traps:**

- *Waterborne vs Vector-borne:* Remember that **Malaria and Dengue are NOT waterborne diseases** (they are insect vector-borne). Cholera, typhoid, hepatitis, and dysentery are waterborne.
- *Primary vs Secondary Air Pollutants:* Smog, Ozone, and PAN are ALWAYS secondary; SO<sub>2</sub>, CO, and NO<sub>x</sub> are primary.
- *Limiting Nutrients:* Freshwater lakes = Phosphorus; Marine systems = Nitrogen.
- *Noise Scale:* The threshold of human hearing is **0 dB** (not 10 or 20 dB).

**4. Pens and Materials:** Use only blue or black ballpoint pen. Do not use gel pens or pencils on OMR sheets.

***ALL THE VERY BEST FOR YOUR EXAMINATION!***

*Review Sections 2, 4, 5, and 6 thoroughly. You have all the knowledge needed to score 90+ marks.*